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Flight Control Systems Hinge Loads Estimation for Aircraft Conceptual Design of Fighters

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Abstract

During aircraft conceptual design, an early evaluation of the on-board systems' configuration has been identified as a promising research topic, with the early definition of power management and configuration leading to cascading effects on the overall design. This is particularly relevant for fighter aircraft, whose aggressive maneuvering requires demanding loads on the flight control system (FCS), and for the ongoing transition toward more-electric architectures, which further increases the need for accurate early-stage load and power estimation. Estimating the overall power required by the FCS is challenging: it depends on actuator loads derived from hinge-moment predictions and typically relies on mathematical correlations involving flight-envelope functions, control-surface geometry, and deflections. Such semi-empirical correlations are fast but assume linear aerodynamics and are poorly suited to the complex three-dimensional geometries of modern fighters, whereas high-fidelity CFD remains too computationally expensive for iterative conceptual design.

The main goal of this thesis is to develop a method to evaluate the hinge loads on a fighter's control surfaces for a specific geometry, along with a 'mission' or 'operating point' defined mainly by altitude and speed, and to perform a comparative analysis between an established semi-empirical approach and a panel-method alternative. The method employs a panel approach, specifically the Tornado MATLAB-based vortex lattice method, integrates various data through the CPACS convention, and is compared against an automated implementation of Roskam's semi-empirical method, both validated against NASA experimental flight data for the F/A-18 SRA.

In the subsonic regime, both methods show good agreement with the flight data, with accuracy degrading toward the transonic region and breaking down at supersonic conditions. Obtaining these hinge moments will ultimately enable better sizing estimations at a low computational cost, compatible with ACD evaluation of preliminary FCS architecture concepts.

Use of Generative AI and AI-Assisted Technologies

The author acknowledges the use of generative AI tools during the preparation of this manuscript. These tools were used to support language editing, including grammar and clarity improvements. To assist in generating selected illustrative figures, and replicate relevant charts or images protected by copyright, citing the source; to help organize the software code structure, and to produce initial draft passages that served as a starting point for further development and revision. All technical content, analyses, interpretations, and conclusions remain the sole responsibility of the author. The author has carefully reviewed, verified, and approved the final version of the manuscript.

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First and foremost, I would like to thank my supervisor, **Erick Espinosa Juárez**, for his constant guidance, availability, and constructive feedback throughout this work. His insight and expertise, combined with his genuine enthusiasm for the topics involved, were fundamental in shaping the direction of this research and in ensuring the validity of the approaches adopted.

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Ringraziamenti

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Ai miei **amici** sparsi in Italia, a chi è sempre rimasto e a chi è passato anche solo per un breve tratto di strada. In questi due anni mi hanno aiutato a non sentirmi solo in un paese inizialmente estraneo, con chiamate costanti e un supporto continuo, sia nei momenti più complessi che in quelli di semplice svago per distrarmi dalla nostalgia. Per due anni è stato come avervi tutti qui con me.

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Per **Sofia**, che è stata il sole in un paese fin troppo buio, un'amica sempre disponibile ad ascoltare i miei problemi e capace di ricordarmi che ogni difficoltà può essere affrontata con molta più leggerezza e serenità di quanto avessi mai immaginato. Dove altri hanno visto una ragazza che non smette mai di parlare, io ho trovato un'amica pronta ad ascoltare e a offrirmi il punto di vista di qualcuno molto più autocritico e introspettivo di quanto non lasci trasparire.

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Tack så mycket!

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Tack så mycket!

1 Introduction

In aircraft conceptual design, early evaluation of on-board systems has emerged as a critical research topic. Traditionally, subsystem definition, particularly for safety-critical systems such as the Flight Control System (FCS), is addressed during later stages of the preliminary design phase. While this approach has proven effective for conventional aircraft configurations derived from extensive historical databases, it becomes increasingly inaccurate when applied to modern and unconventional concepts [1].

The growing interest in more-electric and all-electric aircraft architectures, combined with the limited availability of reliable historical data for novel configurations, has emphasized the importance of system-level design methodologies [2]. Early decisions regarding power management, actuator technology, and redundancy concepts can cascade into mass distribution, thermal management, and overall aircraft performance.

Within this context, estimating the power required by the FCS during the conceptual phase remains particularly challenging. Power consumption depends on actuator loads derived from hinge moment estimations, which are influenced by flight-envelope conditions, control-surface geometry, and aerodynamic interactions. Current empirical methods, such as DATCOM [3] and Roskam [4], provide rapid estimates but are restricted to linear aerodynamic regimes and conventional geometries. They often fail to capture complex 3D interactions and nonlinear effects relevant to modern fighter configurations.

Despite the availability of higher-fidelity CFD approaches, their computational cost makes them unsuitable for iterative conceptual design. Consequently, a methodological gap exists between low-fidelity empirical tools and high-fidelity numerical simulations. There is currently a lack of frameworks capable of linking parametric geometry definition, mission-level performance requirements, hinge moment prediction, and actuator power estimation within a computationally efficient conceptual design environment.

This gap is particularly relevant for fighter aircraft, where complex wing geometries and aggressive maneuvering requirements strongly influence control surface loading and actuation demands. Developing an efficient framework suitable for early-stage fighter design, therefore, represents a relevant research challenge.

1.1 Motivation

Experience in aircraft conceptual design shows that late-stage modifications to the FCS architecture lead to substantial cost increases and schedule delays, potentially requiring a reiteration of early design phases [1]. For safety-critical systems, early architectural decisions regarding actuator technology, redundancy, and power distribution must therefore be supported by reliable estimation tools.

The transition toward more-electric architectures further increases the need for accurate early-stage power and loads estimation. Electric actuator solutions such as Electro-Hydrostatic Actuators (EHA) and Electro-Mechanical Actuators (EMA) offer advantages in terms of power-on-demand and system integration, but they also introduce new challenges related to thermal management, reliability, and peak power requirements [2]. In such architectures, actuator power consumption directly affects generator sizing, electrical distribution systems, and overall aircraft mass.

For fighter aircraft, maneuver-driven loading conditions play a dominant role. Parameters such as altitude, speed, and turn rate significantly alter hinge loads requirements and, therefore, the actuator power requested. However, most preliminary estimation approaches do not integrate mission-specific performance constraints into the actuator sizing process. This motivates the development of a methodology that couples aerodynamic load prediction with mission-level requirements within a conceptual design framework.

1.2 Aim

The main objective of this thesis is to develop a parametric methodology for estimating the hinge moment acting on the ailerons of a fighter aircraft's primary Flight Control System, based on predefined flight condition requirements. Additionally, the study aims to compare results from different estimation methods to assess their consistency and applicability during the conceptual design phase.

The proposed project will:

- Estimate hinge moments based on geometry definition and flight condition using both a semi-empirical and panel method.
- Provide a standardized framework suitable for aircraft conceptual design.
- Enable rapid evaluation of a set configuration within the conceptual design environment, using different estimation methods.

Ultimately, the objective is to support early-stage design decisions by providing a computationally efficient framework to estimate hinge loads and coefficients that could serve as a solid first guess and comparison tool.

1.3 Research questions

Based on the identified methodological gap, the following research questions are formulated:

1. Can a computationally efficient framework combining semi-empirical and panel methods provide sufficient accuracy for hinge moment estimation at the conceptual design stage?
2. Does the integration of a panel method solver improve hinge moment prediction compared to a traditional empirical approach, such as Roskam [4]? And how do they compare?
3. Which flight condition parameters most significantly influence hinge moment prediction accuracy, and do the semi-empirical and panel methods involved respond to them differently?

1.4 Approach

The methodological development is structured in progressive stages. First, a parametric geometry model is implemented in MATLAB and subsequently converted to the CPACS format. Initial hinge-moment estimates are obtained using conventional empirical relations, enabling benchmarking against established preliminary methods.

In a second phase, a low to mid-fidelity aerodynamic solver based on a panel method formulation is integrated. The solver computes pressure distributions over control surfaces, and hinge moments are estimated by summing the loads extracted from each panel. This approach aims to improve geometric sensitivity compared to purely empirical correlations while preserving computational efficiency.

As one of the last steps, a standardized CPACS formatting was implemented for the data in input, translating the previously defined Matlab geometry into a more standardized convention, making the full framework more flexible and compatible with multidisciplinary design optimization.

The assembled tool is intended to generate a lookup table for a given geometry by combining the adopted geometry export convention with the integrated panel method. The resulting database provides hinge-moment estimates across multiple flight conditions using both investigated methodologies (Roskam and Tornado), along with the corresponding aerodynamic results from the panel method.

The resulting data are intended for early-stage design evaluation and, in future developments beyond the scope of this thesis, could be coupled with a flight mechanics model to characterize further the aircraft's aerodynamic behavior under prescribed flight conditions. Within this framework, the FCS sizing process would iteratively use the lookup table to obtain updated hinge-moment estimates at each design point, thereby supporting progressive refinement of aircraft design.

1.5 Delimitation

The scope of this thesis is bounded by a set of deliberate constraints, established to maintain methodological focus and ensure feasibility within the time span of a Master's thesis.

The framework is developed and validated exclusively for a single reference configuration, the F/A-18 Systems Research Aircraft (SRA). While the modular design allows for other geometries to be introduced via the CPACS interface, no additional aircraft configurations are evaluated in this work. The extension of the results to highly swept delta wings, all-moving surfaces, or configurations with non-standard hinge-line geometries cannot be included in the present study without further validation.

The analysis is restricted to the aileron as the primary control surface. Other flight control surfaces, such as rudders, elevators, canards, or leading-edge flaps, are outside the scope of this work, even though the framework could in principle be extended to them.

Both estimation methods employed, Roskam's semi-empirical approach and the Tornado vortex lattice method, are valid only within the linear, attached-flow regime. The study acknowledges but does not resolve the limitations of these methods in the transonic and supersonic regimes. No correction for supersonic aerodynamics is implemented, and results beyond Mach 1 are presented solely to characterize method breakdown rather than to provide usable estimates. Both methods also produce static, steady-state estimates of the hinge moment. Dynamic effects such as rate-dependent aerodynamic loads, actuator lag, and unsteady flow phenomena are not modeled. The reconstruction of dynamic maneuver time histories is performed through lookup-table interpolation under a fixed representative angle of attack, as described in Section 4.3.

The framework is intentionally positioned at the low-to-mid fidelity level, compatible with the iterative evaluation cycles of conceptual design. High-fidelity tools such as Reynolds-Averaged Navier–Stokes (RANS) CFD solvers are excluded based on their computational cost, which is incompatible with the rapid parametric sweeps required at this design stage.

The estimation of actuator power consumption, electrical system sizing, and thermal management, while identified as the broader motivation for this work, falls outside the defined scope. The framework produces hinge-moment estimates that could serve as input to such models in future work, but no actuator or power model is integrated here. CPACS compliance. The CPACS interface implemented in this thesis relies partially on a tool-specific schema extension for parameters that lack a native CPACS equivalent. Full compliance with the official CPACS schema is not achieved and is identified as a future development step.

2 Background and Related Work

The purpose of this chapter is to establish the theoretical and methodological foundations of the thesis and to position the present work within the context of existing research. By reviewing relevant background theory and related scientific contributions, this chapter presents both context-relevant research to contextualize the topic as well as similar tools developed in different environments while highlighting the limitations and advantages.

2.1 Background

This section establishes the theoretical foundations necessary to understand the FCS loads estimation framework.

2.1.1 System-Level Aircraft Conceptual Design

Modern aircraft conceptual design increasingly emphasizes early evaluation of on-board systems and subsystem architectures. Traditionally, subsystem definition, particularly for safety-critical systems such as the Flight Control System (FCS), was addressed in later preliminary design phases. However, recent studies have shown that late architectural changes significantly increase development costs and may require redesign of earlier stages [1].

The shift toward more-electric and all-electric aircraft architectures has underscored the importance of system-level design methodologies that assess subsystem interactions early in the process [2]. In such frameworks, power estimation is no longer a secondary sizing activity but a design driver influencing mass distribution, electrical system sizing, and overall aircraft performance.

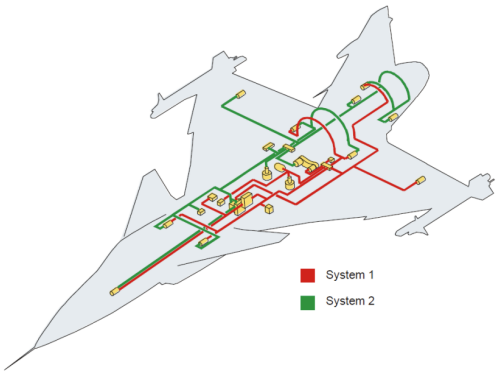
2.1.2 Flight Control Systems and Actuator Technologies

The Flight Control System converts pilot or flight computer commands into control surface deflections, enabling rotations about the aircraft axes [5]. Actuation is typically achieved through:

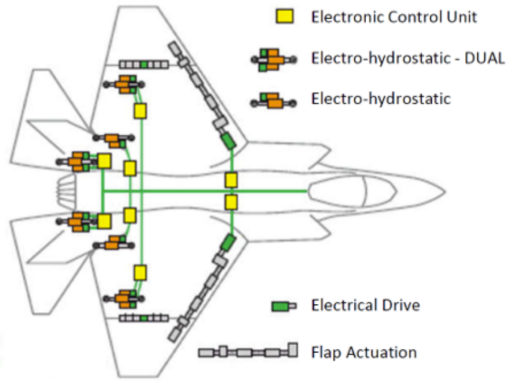
- Servo-Hydraulic Actuators (SHA)
- Electro-Hydrostatic Actuators (EHA)
- Electro-Mechanical Actuators (EMA)

Conventional hydraulic architectures benefit from decades of operational experience and high reliability [6]. However, more-electric configurations, such as those discussed in [2, 7], offer advantages including power-on-demand operation, improved integration with electrical distribution systems, and potential weight reduction.

Examples of hydraulic and more-electric actuation architectures are shown in Figure 2.1.



(a) The Saab Gripen's double redundant hydraulic system. From Landberg [6]



(b) The F-35 actuation system employing Electrohydrostatic Actuators (EHA) and Electromechanical Actuators (EMA) for flap control. From Delbecq [8]

Figure 2.1: Comparison of hydraulic and more-electric actuator system configuration

A schematic overview of the main actuator technologies is shown in Figure 2.2, highlighting architectural differences between hydraulic and electric solutions.

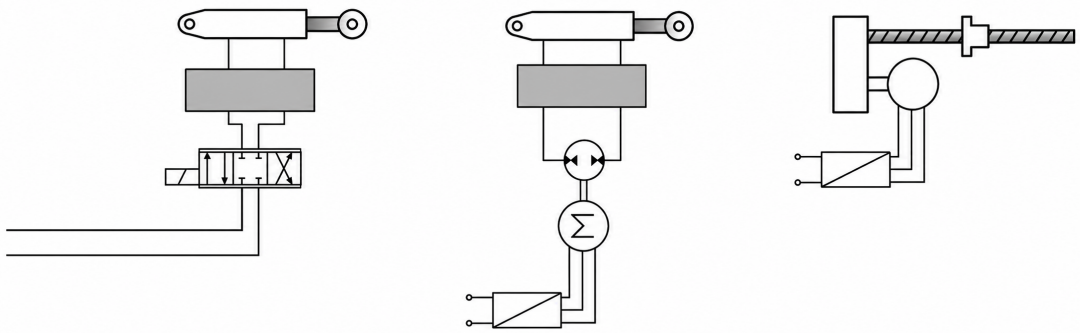


Figure 2.2: Overview of common actuator technologies: Servo-Hydraulic Actuator (SHA), Electro-Hydrostatic Actuator (EHA), and Electro-Mechanical Actuator (EMA) [7].

Electric actuation introduces new challenges related to peak power demand, thermal management, and reliability trade-offs. Consequently, accurate early-stage estimation of actuator loads and power consumption becomes critical in more-electric fighter architectures.

2.1.3 Hinge Moment Theory and Load Estimation

Actuator sizing and power estimation depend fundamentally on hinge-moment prediction. For a control surface, the aerodynamic moment about the hinge line is commonly modeled, within the linear regime, as:

$$M_h = q\, S_f\, c_f\, (C_{h0} + C_{h\alpha}\, \alpha + C_{h\delta}\, \delta) \tag{2.1}$$

where:

M_h	= hinge moment about the hinge line	[Nm]
q	= free-stream dynamic pressure, $q = \frac{1}{2}\rho V^2$	[Pa]
S_f	= control-surface (aileron) reference area	[m^2]
c_f	= control-surface mean aerodynamic chord	[m]
C_{h0}	= baseline hinge-moment coefficient ($\alpha = 0,\ \delta = 0$)	[-]
$C_{h\alpha}$	= derivative of C_h with respect to angle of attack	[rad^{-1}]
α	= angle of attack	[rad]
$C_{h\delta}$	= derivative of C_h with respect to control-surface deflection	[rad^{-1}]
δ	= control-surface deflection	[rad]

The grouping $q\, S_f\, c_f$ represents the dimensional scaling, while the term in brackets is the non-dimensional hinge-moment coefficient $C_h = C_{h0} + C_{h\alpha}\, \alpha + C_{h\delta}\, \delta$, so that $M_h = q\, S_f\, c_f\, C_h$. The baseline term C_{h0} vanishes for a symmetric section at zero incidence and is therefore often negligible, which leaves the two derivatives $C_{h\alpha}$ and $C_{h\delta}$ as the quantities that actually drive the estimate.
These two derivatives are also difficult terms to obtain. Although the model is linear in α and δ, the coefficients are not simple geometric quantities: they depend on the control-surface chord ratio, the nose and aerodynamic-balance shape, the trailing-edge angle, the hinge gap and sealing, the boundary-layer state, the Mach number, and finite-span three-dimensional effects. They also lose physical meaning once the flow separates at a high angle of attack or large deflection, where the assumed linearity breaks down. Estimating $C_{h\alpha}$ and $C_{h\delta}$ reliably is therefore the central problem addressed in this thesis, and their derivation is detailed in Chapter 3.

2.1.4 Empirical Estimation Methods

Empirical methods such as the USAF Stability and Control DATCOM [3] and the methodology proposed by Roskam [4] are widely used in preliminary aircraft design.

DATCOM provides semi-empirical relations for estimating hinge moment derivatives based on trailing-edge geometry assumptions. Its strengths lie in computational simplicity and suitability for iterative conceptual studies. However, it assumes attached flow, linear aerodynamic behavior, and neglects complex three-dimensional interactions [9].

Similarly, Roskam's method offers a structured approach for preliminary control force estimation but relies heavily on historical airfoil data and linear aerodynamic assumptions [4]. Both methods exhibit reduced accuracy when applied to modern or unconventional fighter geometries evaluated in critical conditions.

2.1.5 Panel Methods in Conceptual Aerodynamics

Panel method solvers provide inviscid, potential-flow solutions with significantly lower computational cost than full CFD simulations. While they neglect viscous and separated flow effects, they offer improved geometric sensitivity compared to purely empirical methods.

Due to their computational efficiency, panel methods are commonly considered suitable for conceptual design environments that require rapid iteration. Their potential application to hinge moment estimation represents a mid-fidelity alternative between empirical correlations and high-fidelity CFD approaches.

Tornado

Tornado is a three-dimensional vortex lattice method implemented in MATLAB. It is primarily designed for the conceptual design stage of aircraft development and serves as an effective tool in the early evaluation of different concepts. The software computes 3D forces acting on individual wing panels, global aerodynamic coefficients, and stability derivatives with respect to the angle of attack, sideslip, angular rates, and control surface deflections. Tornado features a text-based menu system for navigating geometry setup, flight state definition, and post-processing tasks.

The program is based on standard vortex lattice theory derived from potential flow theory. A defining characteristic of Tornado is its flexible wake model; unlike traditional rigid wake models, the wake trailing from the lifting surfaces varies with flight conditions. Furthermore, the software replaces the classical three-segment "horse-shoe" vortex with a seven-segment "vortex-sling" arrangement. To determine the first-order stability derivatives, Tornado uses a central difference calculation by disturbing a pre-selected flight state by a small amount.

Its primary advantages include a highly flexible design system that supports multi-wing configurations with complex features like sweep, taper, camber, and various control surfaces (e.g., flaps, ailerons, and canards). A significant technical strength is its use of a dynamic "flexible wake" and a seven-segment "vortex-sling" arrangement, which allows the program to capture certain non-linear effects, such as main wing and stabilizer interactions, more accurately than traditional rigid-wake models. Furthermore, its open-source nature under the GNU license and its MATLAB-based architecture ensure high portability across different operating systems. In contrast, its disadvantages stem largely from the inherent limitations of vortex lattice theory, which assumes small angles of attack and neglects critical physical factors such as fuselage effects, friction drag, air compressibility, and the physical thickness of lifting surfaces.[10]

2.2 Related Work

This section reviews scientific contributions related to early FCS estimation, actuator system comparison, and hinge moment prediction methodologies.

2.2.1 Early Estimation of FCS Mass and Power

Recent research emphasizes integrating mass and power estimation into early FCS design phases. Lübbe et al. [1] demonstrate that early architectural decisions significantly influence aircraft-level mass and power budgets, advocating integrated estimation methodologies during preliminary design.

Chakraborty et al. [2] propose a vehicle- and mission-level comparison framework for more-electric actuator systems, highlighting the impact of subsystem selection on aircraft performance and energy demand.

2.2.2 Comparison of Hydraulic and Electric Actuation Architectures

Comparative studies between hydraulic and electric actuation systems analyze trade-offs in reliability, efficiency, weight, and power distribution. Dell'Amico and Reichenwallner [7] provide a conceptual comparison tailored to fighter aircraft, emphasizing integration challenges and power density considerations.

These studies underscore the need for accurate actuator load and power predictions in early design phases, particularly for more-electric configurations.

2.2.3 Modeling and Simulation of Hinge Moments

The development of simulation environments for the performance of electric actuators under maneuvering flight conditions has been investigated by Chakraborty et al. [11]. Their work links aerodynamic loading to actuator behavior in dynamic conditions.

CFD-based hinge moment prediction methods, such as those investigated by Simpson [9], demonstrate improved accuracy under complex flow conditions but at significantly higher computational cost.

These studies illustrate the trade-off between fidelity and computational efficiency, motivating the investigation of mid-fidelity approaches.

2.2.4 Limitations in Existing Approaches

Existing empirical methods are limited by linear aerodynamic assumptions and reduced geometric sensitivity, while high-fidelity CFD approaches are computationally expensive and unsuitable for iterative conceptual design.

Although previous works address actuator comparison, hinge moment modeling, and subsystem-level estimation separately, there is limited integration of:

- Parametric geometry modeling,
- Mission-driven aerodynamic loading,
- Mid-fidelity hinge moment prediction,
- Application to fighter conceptual design.

This integration gap directly motivates the methodological development proposed in this thesis.

2.3 Design Choices - Prestudy

Considering previous studies and the overall aim of this thesis, an envisioned potential framework is illustrated in Fig. 2.3, including future implementations that may follow this work, such as a flight-mechanics model and couplings with design tools. This would refine the performance requirements based on the aircraft's actual behavior during prescribed maneuvers, as well as actuator models that estimate the effective power required to operate the ailerons with electric actuators under defined conditions.

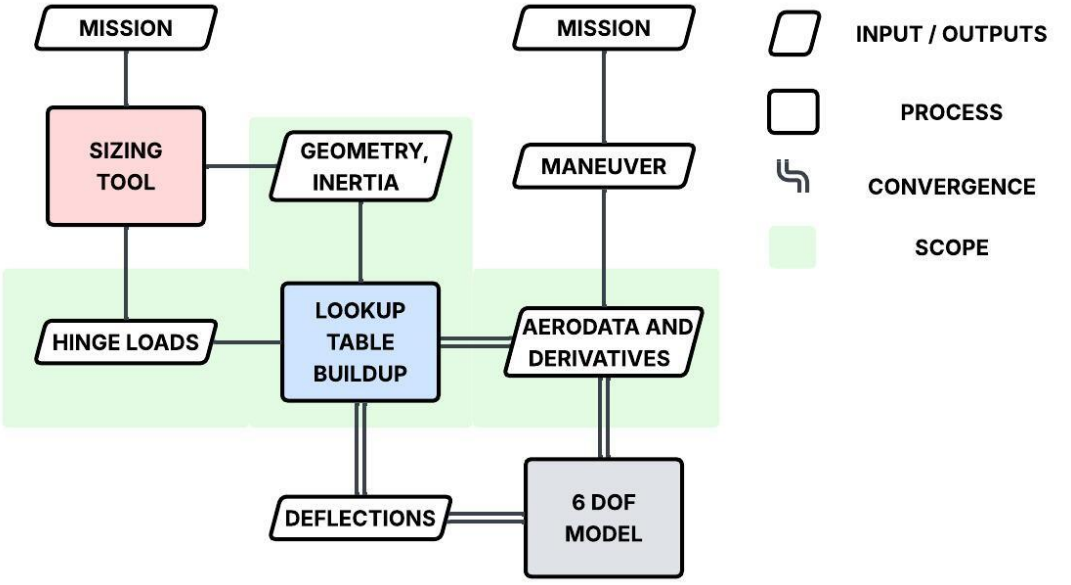


Figure 2.3: Complete Framework with Out of scope elements to illustrate complete context in an XSDM formatting, with inputs coming from the top and bottom sides of each square (or process) and outputs on the sides, leading to converging iterations highlighted by the double connections

The scope of the thesis highlighted in green is composed by different modules that will first be introduced from a theoretical perspective in the upcoming sections 2.4-2.5, and later their respective implementation will also be described in Chapter 3 (sections 3.1-3.2).

The final scope of the thesis is, in fact, focused on the objective of creating a modular and standardized framework that can be used for the evaluation of early concept designs, predicting hinge loads under specific flight condition requirements, and initiating a link between standardized CPACS geometry definition exported directly from design sizing tools, and both empirical and panel methods, enabling comparison of the results obtained with different approaches.

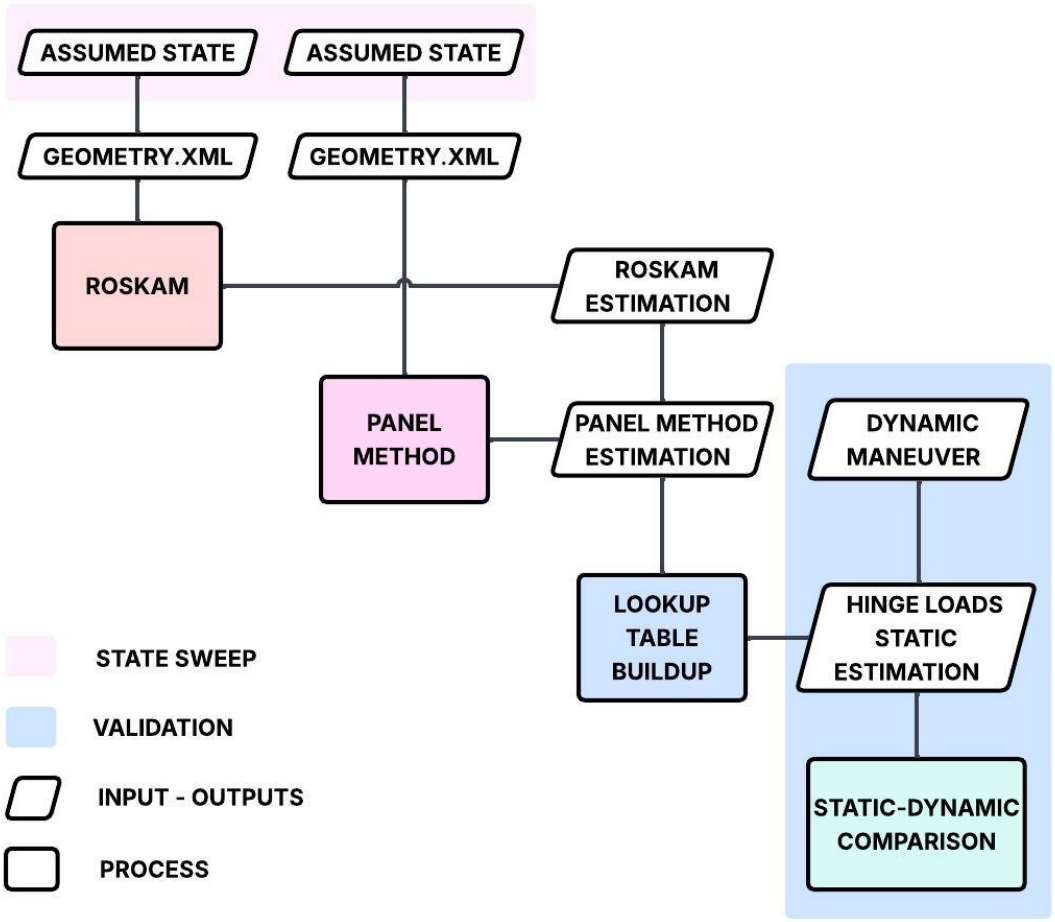


Figure 2.4: Modular methodology adopted in this thesis, within the defined scope in an XSDM formatting, with inputs coming from the top and bottom sides of each square (or process) and outputs on the sides, highlighting the flight condition (state) sweep as well as the final comparison from a dynamic maneuver.

The final in-scope methodology in Fig.2.4 can be divided into several sections approached individually and successively merged.
The first section concerns geometric data storage, which, in future studies, might be derived from design software and tools in the CPACS convention, but was defined preliminarily in MATLAB using a similar nomenclature and later converted to CPACS.

The second section presents an automated MATLAB implementation of semi-empirical methods, such as Roskam, that provides a first estimate of hinge loads and coefficients under specified geometry and flight-condition requirements.

A third section is defined by a panel code that aims to obtain alternative estimates of hinge loads and coefficients using a VLM method, which can be compared with the previous results and stored in a lookup table for the set conditions together with semi-empirical derivations. The table mentioned also collects aerodynamic results useful for obtaining flight control derivatives that, in future studies, may be imported into a flight mechanics model to enhance performance requirements.

The results of both methodologies are finally organized into a lookup-table structure for each evaluated condition after looping over different flight setups. A schematic of the final output

is shown in Fig.2.5 and was finally used to validate the static results obtained through state iterations, with real flight data test and simulations further described in Ch.4.

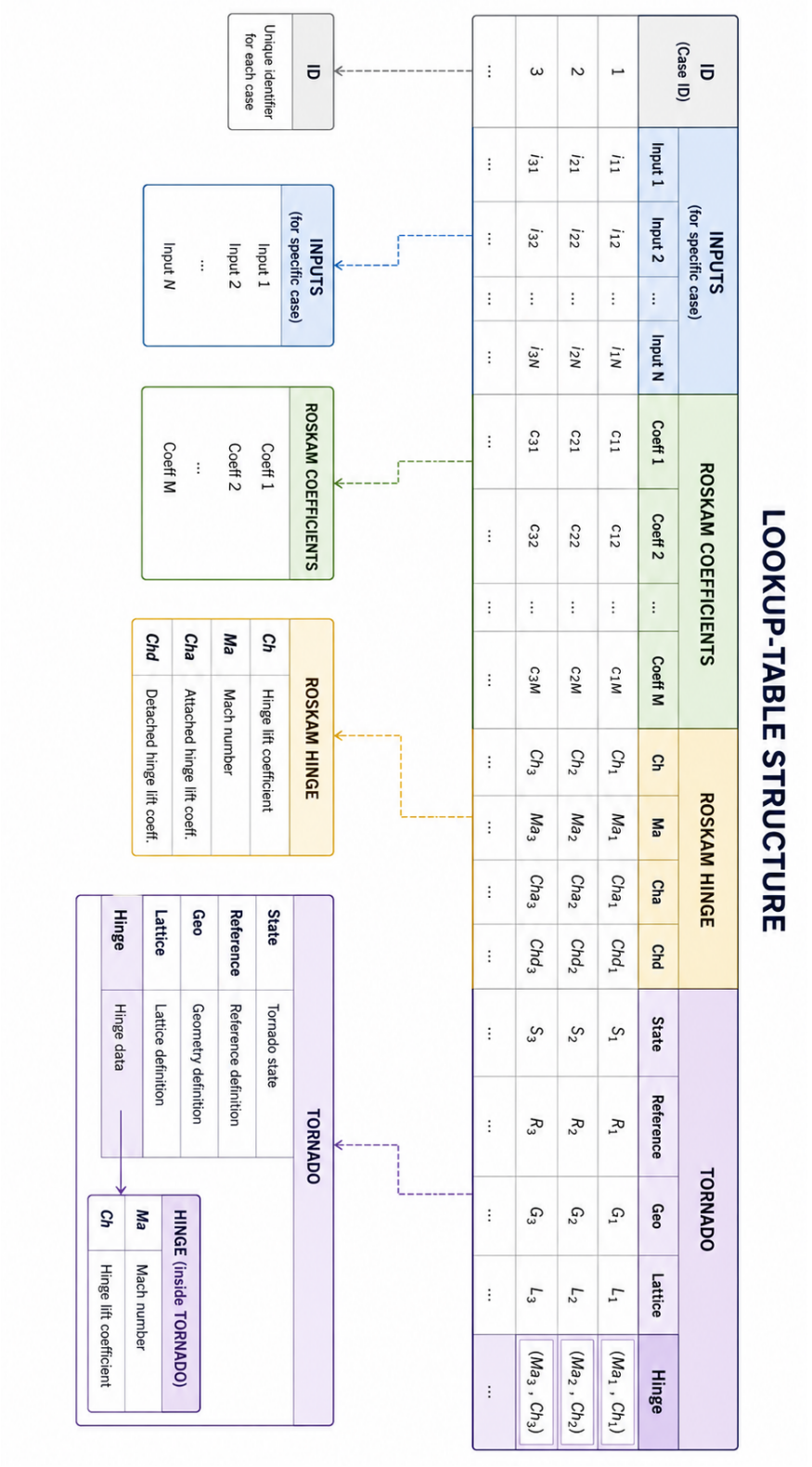


Figure 2.5: Schematic Representation of the final Lookup Table in output

2.4 Roskam

The Roskam methodology serves as a robust "first-guess" tool for preliminary design, yet its accuracy is strictly confined by its empirical foundations. Because it relies heavily on historical NACA wind-tunnel data, it assumes a linear relationship among the hinge-moment coefficient (C_h), the angle of attack, and the control-surface deflection. This linearity collapses during high-alpha (above 12°) maneuvers or large deflections (around 20°) where flow separation and non-linear boundary layer effects dominate.

Furthermore, the method struggles with modern aerodynamic nuances, such as transonic flow, where shock-wave interactions shift pressure distributions, and with sensitive geometric details, such as seal types and gap sizes. While it provides a reliable baseline for initial sizing, it lacks the high-fidelity precision required to validate modern, non-conventional profiles or fly-by-wire actuators without supplementary CFD or wind-tunnel validation.
Figure 2.6 illustrates the Roskam implementation as a black-box model, emphasizing the relationship between the input parameters and the resulting outputs.

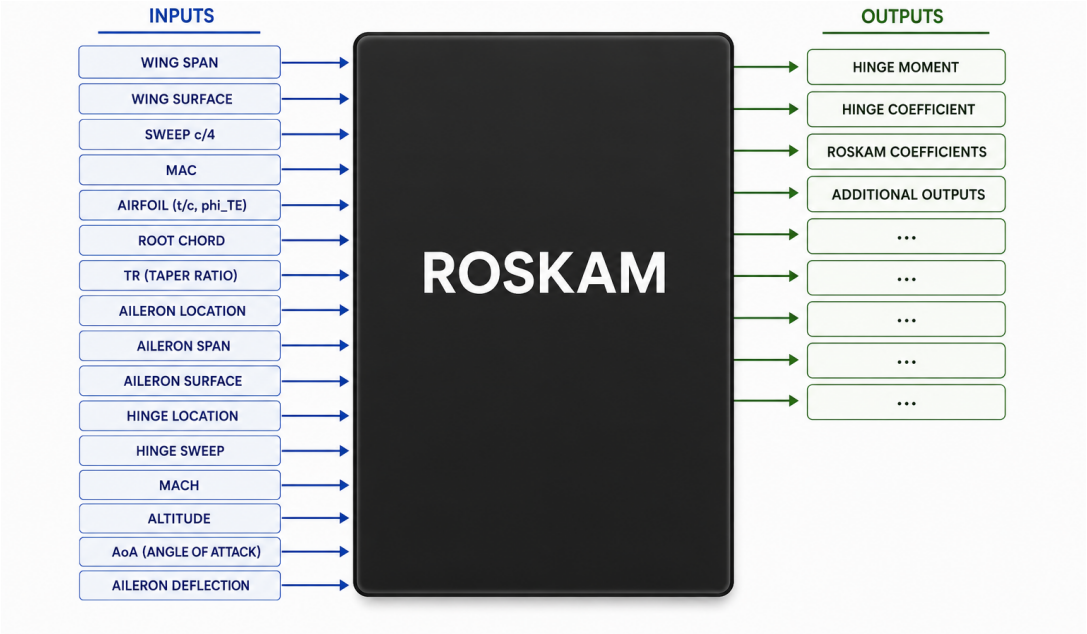


Figure 2.6: Black Box representation of Roskam Method

Chapter 3.1 details the implementation in MATLAB of the Roskam empirical methodology used to estimate hinge loads on the aileron of a defined geometry at a set flight condition.

2.5 Panel Methods

Panel methods are a class of potential-flow numerical techniques used in aerodynamics to model the flow around bodies by discretization of their surfaces into a finite number of panels, on which singularity distributions (such as sources, doublets, or vortices) are applied.

These methods assume inviscid, incompressible, and irrotational flow, making them computationally efficient while still capturing key aerodynamic characteristics. Among the most widely used panel methods is the Vortex Lattice Method (VLM), which represents lifting surfaces using a lattice of bound vortices and is particularly effective for analyzing wings and high-aspect-ratio configurations. An example of a VLM-based tool is TORNADO, which is commonly used in preliminary aircraft design for rapid estimation of lift distribution, induced drag, and stability derivatives. The black-box model of the Panel method used in this thesis is shown in Fig. 2.7 and will be detailed in the following chapters.

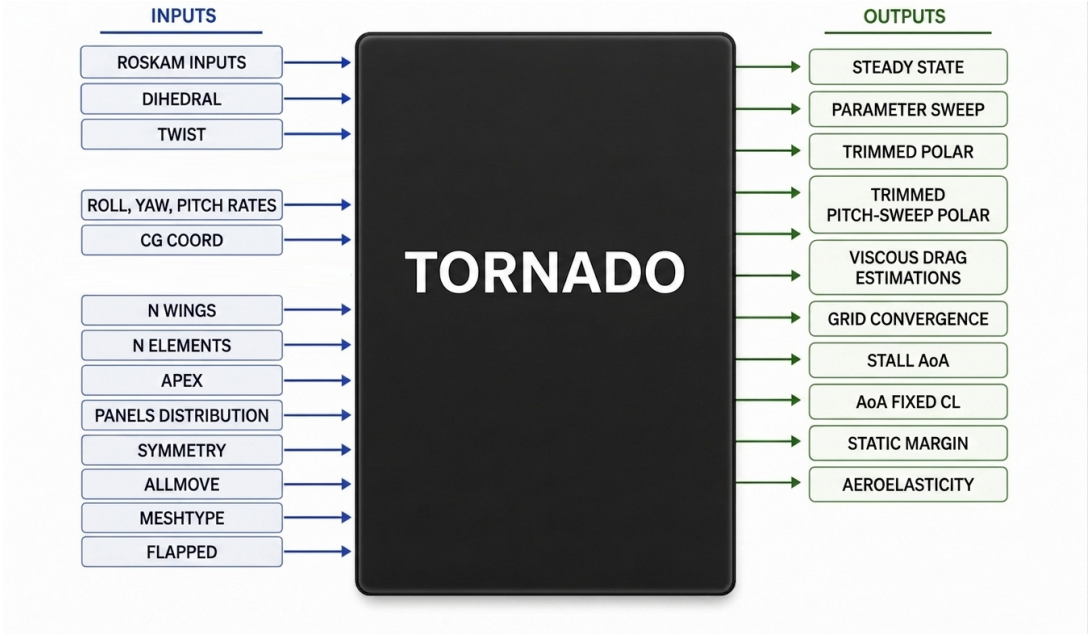


Figure 2.7: Black Box representation of Tornado Panel Method

The main advantages of panel methods include their low computational cost, ease of implementation, and suitability for early-stage design and parametric studies. However, their limitations stem from underlying assumptions: they neglect viscous effects, flow separation, and boundary-layer development, and they provide limited accuracy near sharp edges or in highly nonlinear flow regimes.

Consequently, while panel methods are valuable for conceptual design and initial analysis, they must often be complemented by higher-fidelity tools or empirical corrections for detailed aerodynamic assessments. They can be seriously inaccurate in extreme conditions to which a fighter might be subjected. Therefore, they should be used as a comparison tool that highlights the speed with which they can provide a lot of data from very simple input geometries, and not trusted entirely regarding their accuracy, which higher-level CFD analysis remains superior.

3 Methodology And Implementation

3.1 MATLAB automation of Roskam's Method

3.1.1 From the Hinge-Moment Model to Coefficient Estimation

The hinge-moment model adopted in this work was introduced in Section 2.1.3 (Eq. 2.1). The dimensional scaling $q\,S_f\,c_f$ follows directly from the flight condition and the control-surface geometry, so the estimation problem reduces to determining the two hinge-moment derivatives $C_{h\alpha}$ and $C_{h\delta}$. The remainder of this section describes how these coefficients are obtained through Roskam's semi-empirical correction chain, using the geometric definitions illustrated in Figure 3.1.

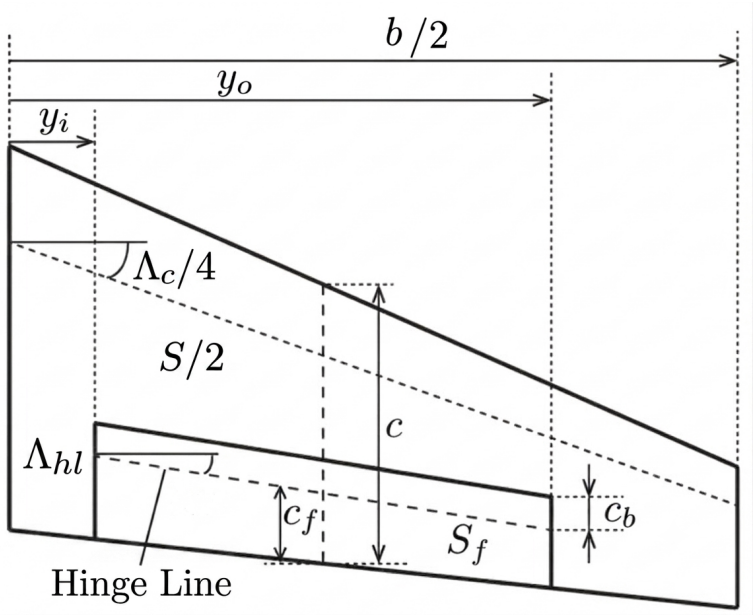


Figure 3.1: Control-surface geometry and hinge-line definition for hinge-moment estimation with updated correct definition of c_f from trailing edge until hinge line. [11]

3.1.2 Code Architecture

To provide a clearer overview of Roskam's semi-empirical estimation method, a MATLAB code can be developed to digitize and accelerate the process, enabling its application to different geometries and flight conditions.

The overall MATLAB code architecture is shown in Fig.3.2, divided into its main scripts and function components. A full picture of the complete framework is also available in Appendix 8.1, including this integration and the upcoming steps.

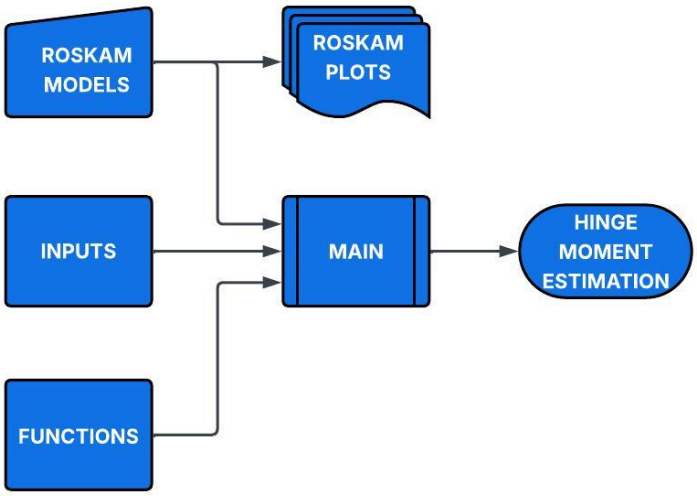
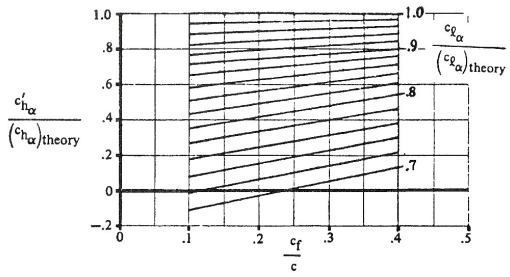


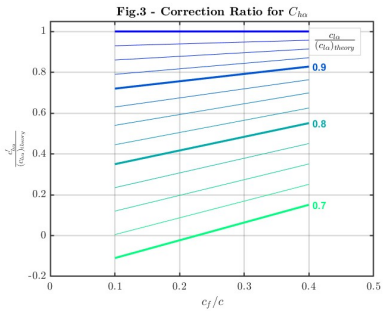
Figure 3.2: Overall MATLAB Code architecture for Roskam Methodology implementation

3.1.3 Roskam Models and Plots

A particularly relevant aspect of Roskam's methodology, described in Volume VI [4], is the extrapolation of different component values and coefficients from experimental plots. A series of relevant graphs is digitized using MATLAB functions, which will later be used to extrapolate the required values, and plotted for visual inspection. An example case is shown in Fig.3.3.



(a) Reproduction of Roskam Plot for $C'_{h\alpha}$ correction ratio found in [4] (Page 468)



(b) MATLAB digitalized version of $C'_{h\alpha}$ correction ratio

Figure 3.3: Example of a digitization of the original plot from Roskam and the version created on MATLAB

3.1.4 Main Script and inputs

A main script manages the full code architecture, initially using only an aircraft-specific ID to select the geometry and the corresponding previously uploaded airfoil details. This modularity allows for the easy introduction of new geometries and flight conditions, initially used to check the ongoing implementation. The initially loaded inputs included two geometries as sample data (F-35, F/A-18 SR) associated with the same airfoil (NACA 64A210) but with different flight conditions used to evaluate the results.

A summary table of the inputs needed for the full evaluation is presented in Table.3.1.

Variable	Description	Unit
	— Geometry —	
b	Wing span	m
S	Wing reference area	m^2
c_r	Root Chord	m
c_t	Tip Chord	m
Λ	Wing sweep angle	deg
$\Lambda_{\frac{c}{4}}$	Wing sweep angle at 1/4 of chord	deg
Γ	Wing dihedral angle	deg
y_i	Inboard spanwise location of aileron	m
y_o	Outboard spanwise location of aileron	m
S_f	Aileron surface area	m^2
Hinge position	Aileron hinge location as percentage of chord	$\%c_f$
	— Flight Conditions —	
M	Flight Mach number	–
Altitude	Geopotential altitude	m
α	Angle of attack	deg
δ	Aileron deflection angle	deg
	— Airfoil Data —	
t/c	Airfoil thickness-to-chord ratio	–
C_{l_α}	Airfoil lift curve slope (2D)	1/rad
Point coordinates	Normalized (x, z) airfoil coordinates	–

Table 3.1: Input Parameters for MATLAB Hinge Estimation Model

3.1.5 Functions

Auxiliary functions

A series of auxiliary functions is also used mainly to obtain derived data from the inputs presented that will later become useful, especially all those values that, like geometrical dimensions, might be derived from fewer inputs. Three functions are then used to obtain derived wing and airfoil geometry values from given inputs, as well as flight conditions such as air properties for the set altitude and Mach number.

Main Functions

After all inputs and useful functions are fully set, the two main functions are finally used to obtain the estimations. The following subsections will illustrate the full Roskam's full methodology implemented step by step.

Derivation of C_{h_α}

The first coefficient C_{h_α} is the hinge-moment derivative with respect to the angle of attack (also known as *floating derivative*), which can be obtained from Eq.3.1, is composed of three main components; the first one in brackets can be easily obtained with geometrical dimensions from the wing definition.

$$C_{h_\alpha} = \left(\frac{AR \cos \Lambda_{c/4}}{AR + 2 \cos \Lambda_{c/4}} \right) C_{h_{\alpha,M}} + \Delta C_{h_\alpha} \tag{3.1}$$

where:

$C_{h\alpha}$ = hinge-moment derivative w.r.t. angle of attack (*floating derivative*) $[rad^{-1}]$
$C_{h_{\alpha,M}}$ = Hinge moment derivative corrected for Mach number effect
AR = Aspect ratio of the wing
$\Lambda_{c/4}$ = Sweep angle at the quarter-chord line
ΔC_{h_α} = Induced effect due to spanwise lift distribution

Mach Number Correction Component

The second component of Eq.3.1 ($C_{h_{\alpha,M}}$) is calculated using Eq.3.2.

$$C_{h_{\alpha,M}} = \frac{(C_{h_\alpha})_{balance}}{\sqrt{1 - M^2}} \tag{3.2}$$

where:

$C_{h_{\alpha,M}}$ = Mach-corrected hinge moment derivative
$(C_{h_\alpha})_{\text{balance}}$ = Hinge moment derivative corrected for aerodynamic balance
M = Flight Mach number

Where the main component of the derivative coefficient corrected for balance can be decomposed as follows:

$$(C_{h_\alpha})_{balance} = C''_{h_\alpha} \frac{(C_{h_\alpha})_{balance}}{C''_{h_\alpha}} \tag{3.3}$$

where:

C''_{h_α} = Hinge moment derivative corrected for thickness
$(C_{h_\alpha})_{\text{balance}}$ = Hinge moment derivative corrected for aerodynamic balance

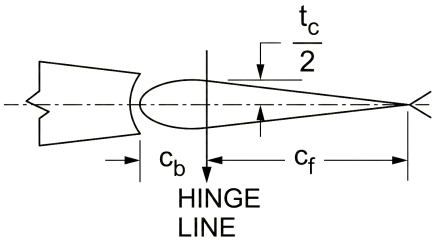
The ratio highlighted in Eq.3.3 can now be extracted from the plot shown in Roskam (*Chapter 10, Fig-10.65a, p.471*), using the nose shape and the Balance Ratio, calculated as in Eq.3.4, as inputs for the plot.

$$BR = \sqrt{\left(\frac{c_b}{c_f} \right)^2 - \left(\frac{t_c}{2c_f} \right)^2} \tag{3.4}$$

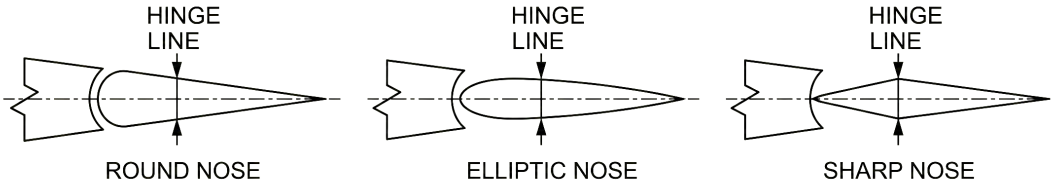
where:

c_b/c_f = Overhang ratio
BR = Balance ratio
t_c = Absolute thickness of the control surface at the hinge line

Which takes into account the nomenclature followed by Roskam and shown in Fig.3.4.



(a) Control surface definitions



(b) Nose-types classifications for control surface

Figure 3.4: Reproductions of Roskam definitions for Balance Ratio corrections (*Roskam VI Chapter 10 p.471*)

To complete Eq.3.3, one more component is needed (C''_{h_α}). At this point, the condition shown in Eq.3.5 is verified considering the trailing edge geometry of the control surface as defined in Fig.3.5.

$$\frac{t}{c} = tg(\frac{\Phi_{TE}}{2}) = tg(\frac{\Phi'_{TE}}{2}) = tg(\frac{\Phi''_{TE}}{2}) \tag{3.5}$$

where:

t/c = Airfoil thickness-to-chord ratio
Φ_{TE} = Trailing-edge angle
Φ'_{TE} = Trailing-edge angle computed between 0.90 and 0.99 c
Φ''_{TE} = Trailing-edge angle computed between 0.95 and 0.99 c

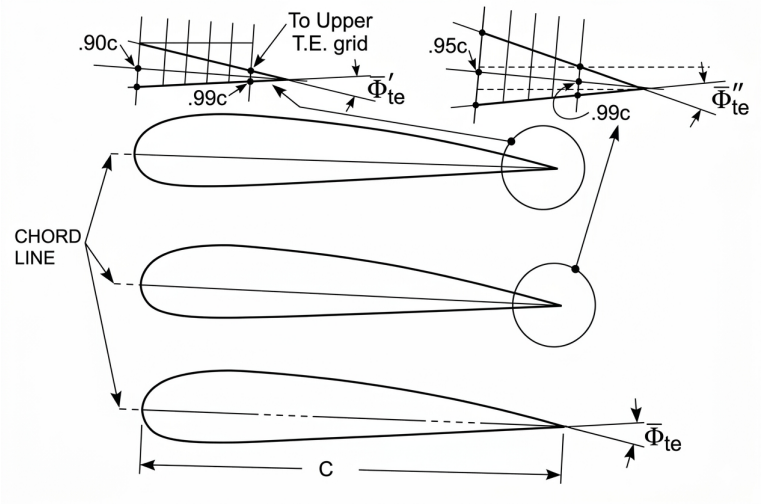


Figure 3.5: Reproduction of Trailing Edge Angles definition (*Roskam VI Ch-10, p.465, fig.10.62*)

In the code, this verification is handled by checking if the mean value between the two angle approximations verifies the condition with an adjustable tolerance set (fixed by default at 10^{-2}).
If the condition expressed in Eq.3.5 is satisfied then the coefficient previously needed to complete Eq.3.3 can be obtained as follows:

$$C''_{h_\alpha} = C'_{h_\alpha} = \left(\frac{C'_{h_\alpha}}{C_{h_{\alpha(\text{theory})}}} \right) \cdot C_{h_{\alpha(\text{theory})}} \tag{3.6}$$

where:

$$\begin{aligned} C''_{h_\alpha} &= \text{Hinge moment derivative corrected for thickness} \\ C'_{h_\alpha} &= \text{Hinge moment derivative corrected for viscosity} \\ C_{h_{\alpha(\text{theory})}} &= \text{Theoretical 2D hinge moment derivative} \end{aligned}$$

The two main components constituting this expression can be extracted from Roskam plots using the functions previously realized. Specifically, the ratio highlighted in the brackets in Eq.3.6 can be extrapolated from the plot shown in Roskam (*Chapter 10, Fig.10.63a, p.468*) using the Aileron chord ratio ($\frac{c_f}{c}$) calculated with respect to MAC (Mean Aerodynamic Chord of the wing). A second input is needed to obtain the correct value from this plot: the ratio between the real Lift-Angle of attack (dependent on flight conditions, eg, Re), curve slope, and the theoretical value for it. These two components are respectively extracted from the plots in Roskam VI (*Chapter 10 Fig.10.64a-b, p.469*), where the tangent of half the angle Φ'_{TE} is used, coupled with the Reynolds number calculated with respect to the MAC; while the theoretical value is extracted form the other plot using the thickness ratio (t/c) as only input. With this data, the coefficients in Eq.3.6 can be obtained.

If the condition expressed in Eq.3.5 is not satisfied, on the other hand, then the coefficient needed to complete Eq.3.3 can be calculated as follows:

$$C''_{h_\alpha} = C'_{h_\alpha} + 2C_{L_{\alpha(theory)}}(1 - \frac{C_{L_\alpha}}{C_{L_{\alpha,\text{theory}}}}) \left[\tan\left(\frac{\Phi''_{TE}}{2}\right) - (t/c)\right] \tag{3.7}$$

where:

C''_{h_α} = Hinge moment derivative corrected for thickness
C'_{h_α} = Hinge moment derivative corrected for viscosity
C_{L_α} = Real lift-curve slope of the airfoil
$C_{L_{\alpha,\text{theory}}}$ = Theoretical lift-curve slope
Φ''_{TE} = Trailing-edge angle computed between 0.95 and 0.99 c
t/c = Wing thickness-to-chord ratio

At this point, all components are known and the coefficient for the Mach correction, expressed in Eq.3.2, can finally be obtained.

3D Hinge Coefficient Component

The final component needed to complete Eq.3.1 is ΔC_{h_α}, and can be obtained with the following decomposition:

$$\Delta C_{h_\alpha} = (\frac{\Delta C_{h_\alpha}}{C_{L,\alpha} B_2 K_a \cos(\Lambda_{c/4})})\, C_{L,\alpha}\, B_2\, K_a\, \cos(\Lambda_{c/4}) \tag{3.8}$$

where:

ΔC_{h_α} = Increment of hinge-moment corrected for 3D effects
C_{L_α} = Lift-curve slope
B_2 = Geometric correction coefficient
K_a = Mean aerodynamic influence coefficient
$\Lambda_{c/4}$ = Quarter-chord sweep angle

The first term, in the brackets, at the beginning of Eq.3.8 can be easily extracted from Roskam VI (*Fig.10.77a, Ch-10, p.483*) using the wing's aspect ratio (AR).

Form geometrical data, the sweep angle at a quarter of the chord can be extracted; the slope coefficient C_{L_α} is also known after multiplying the ratio previously extracted to complete Eq.3.7 by the previously obtained theoretical value, as well as for the same equation.

The term B_2, which accounts for the control surface and balance chord ratio, can be extracted from Roskam VI (*Fig.10.77c, Ch-10, p.483*) using the overhang ratio (c_b/c_f) and the chord ratio ($\frac{c_f}{c}$) previously used.

The last component needed is K_a, which can be obtained from Eq.3.9 after extrapolating the needed coefficients (K_{ai} and K_{ao} from the corresponding plot in Roskam VI (*Fig.10.77b, Ch-10, p.483*), using the geometrical definitions reported in Fig.3.1.

$$K_a = \frac{K_{ai}(1-\eta_i) - K_{ao}(1-\eta_o)}{\eta_o - \eta_i} \tag{3.9}$$

where:

$$\begin{aligned} K_a &= \text{Mean aerodynamic influence coefficient} \\ K_{ai} &= \text{Inner aerodynamic influence coefficient} \\ K_{ao} &= \text{Outer aerodynamic influence coefficient} \\ \eta_i &= \text{Inner non-dimensional spanwise location} \\ \eta_o &= \text{Outer non-dimensional spanwise location} \end{aligned}$$

Finally, ΔC_{h_α} can be obtained as well as the full estimation for C_{h_α} can be obtained using Eq.3.1.

Derivation of C_{h_δ}

The derivation of this second coefficient, the hinge moment derivative with respect to the aileron's deflection, starts with this formulation:

$$C_{h_\delta} = \cos\Lambda_{c/4}\cos\Lambda_{hl}\left(C_{h_{\delta,M}} + \alpha_\delta C_{h_{\alpha,M}}\frac{2\cos\Lambda_{c/4}}{AR + 2\cos\Lambda_{c/4}} + \Delta C_{h_\delta}\right) \tag{3.10}$$

where:

$$\begin{aligned} C_{h\delta} &= \text{hinge-moment derivative w.r.t. deflection (}\textit{restoring derivative}\text{)} \quad [rad^{-1}] \\ C_{h_{\delta,M}} &= \text{Mach/corrected hinge moment derivative affected by aileron deflection} \\ \Lambda_{c/4} &= \text{Quarter of wing chord sweep} \\ \Lambda_{hl} &= \text{Hinge-line sweep angle} \\ \alpha_\delta &= \text{Flap effectiveness parameter} \\ \Delta C_{h_\delta} &= \text{3D component due to control surface deflection} \\ AR &= \text{Wing Aspect Ratio} \end{aligned}$$

Using the wing's geometrical information, many components are already known ($\Lambda_{c/4}$, Λ_{hl}, AR), leaving two major components to be derived ($C_{h_{\delta,M}}$ and ΔC_{h_δ}), respectively the Mach number and 3D corrections.

Mach Number Correction Component

Similarly to what was previously done for the angle alpha component, the Mach correction can be obtained as follows:

$$C_{h_{\delta,M}} = \frac{(C_{h_\delta})_{balance}}{\sqrt{1-M^2}} \tag{3.11}$$

where:

$$\begin{aligned} C_{h_{\delta,M}} &= \text{Mach-corrected hinge moment derivative} \\ \left(C_{h_\delta}\right)_{\text{balance}} &= \text{Hinge moment derivative corrected for aerodynamic balance} \\ M &= \text{Flight Mach number} \end{aligned}$$

Once again, the main component of the derivative coefficient corrected for balance can be decomposed as follows:

$$(C_{h_\delta})_{balance} = C''_{h_\delta} \frac{(C_{h_\delta})_{balance}}{C''_{h_\delta}} \tag{3.12}$$

where:

$$C''_{h_\delta} = \text{Hinge moment derivative corrected for thickness}$$
$$(C_{h_\delta})_{\text{balance}} = \text{Hinge moment derivative corrected for aerodynamic balance}$$

The ratio highlighted in Eq.3.12 can now be extracted from the plot shown in Roskam VI(*Chapter 10, Fig-10.71, p.477*), using the nose shape and the Balance Ratio, calculated as in Eq.3.4, as inputs for the plot.

To obtain the component C''_{h_δ} the came condition proposed in Eq.3.5 is once again used in the exact same way, and if satisfied, the following applies:

$$C''_{h_\delta} = C'_{h_\delta} = \left(\frac{C'_{h_\delta}}{C_{h_{\delta(\text{theory})}}} \right) \cdot C_{h_{\delta(\text{theory})}} \tag{3.13}$$

where:

$$C''_{h_\delta} = \text{Hinge moment derivative corrected for thickness}$$
$$C'_{h_\delta} = \text{Hinge moment derivative corrected for viscosity}$$
$$C_{h_{\delta(\text{theory})}} = \text{Theoretical 2D hinge moment derivative}$$

The two main components constituting this expression can be extracted from Roskam plots once more. Specifically, the ratio highlighted in the brackets in Eq.3.13 can be extrapolated from the plot shown in Roskam (*Chapter 10, Fig.10.69a, p.475*) using the Aileron chord ratio ($\frac{c_f}{c}$) calculated with respect to MAC (Mean Aerodynamic Chord of the wing). Again, the second input is needed to obtain the correct value from this plot: the ratio between the real Lift-Angle of attack (dependent on flight conditions, eg, Re), curve slope, and the theoretical value for it, which was previously obtained (*Chapter 10 Fig.10.64a-b, p.469*). With this data, the coefficients in Eq.3.13 can be obtained.

Similarly, if the condition expressed in Eq.3.5 is not satisfied, then the coefficient needed to complete Eq.3.3 can be calculated as follows:

$$C''_{h_\delta} = C'_{h_\delta} + 2C_{L_{\delta(theory)}} (1 - \frac{C_{L_\delta}}{C_{L_{\delta,\text{theory}}}}) \left[\tan\left(\frac{\Phi''_{TE}}{2} \right) - (t/c) \right] \tag{3.14}$$

where:

$$C''_{h_\delta} = \text{Hinge moment derivative corrected for thickness}$$
$$C'_{h_\delta} = \text{Hinge moment derivative corrected for viscosity}$$
$$C_{L_\delta} = \text{Real lift-curve slope of the airfoil}$$
$$C_{L_{\delta,\text{theory}}} = \text{Theoretical lift-curve slope}$$
$$\Phi''_{TE} = \text{Trailing-edge angle computed between 0.95 and 0.99 c}$$
$$t/c = \text{Wing thickness-to-chord ratio}$$

But this time the values for $C_{L_{\delta,\text{theory}}}$ and C_{L_δ} need to be extracted from Roskam VI (*Fig.8.14-8.15 Ch-8, p,228-229*), using as inputs respectively the chord (c_f/c) and thickness (t/c) ratios paired with the slope coefficient ratio ($C_{L,\alpha}/C_{L,\alpha(theory)}$) previously obtained.

Finally, the Mach-corrected component can be obtained for the deflection angle of the aileron with Eq.3.11.

3D Component

A similar procedure as the one previously used for the 3D effect on the coefficient $C_{h,\alpha}$ is also used in this case to obtain ΔC_{h_δ}.

$$\Delta C_{h_\delta} = \left(\frac{\Delta C_{h_\delta}}{C_{L,\delta} B_2 K_d \cos(\Lambda_{c/4})}\right) C_{L,\delta}\, B_2\, K_a\, \cos(\Lambda_{c/4}) \tag{3.15}$$

where:

$$\begin{aligned}
\Delta C_{h_\alpha} &= \text{Increment of hinge-moment corrected for 3D effects} \\
C_{L_\alpha} &= \text{Lift-curve slope} \\
B_2 &= \text{Geometric correction coefficient} \\
K_d &= \text{Mean aerodynamic influence coefficient for deflection angle} \\
\Lambda_{c/4} &= \text{Quarter-chord sweep angle}
\end{aligned}$$

The highlighted term in the brackets at the beginning of the equation can be easily extracted once more from Roskam VI (*Fig.10.78a, Ch-10, p.485*) using the wing's aspect ratio (AR).

Other than the geometrical data (assumed known at this point), the slope coefficient C_{L_δ} is also computed by multiplying the ratio previously extracted to complete Eq.3.14 by the previously obtained theoretical value, as well as for the same equation.

The term B_2 remains valid in this case as well (obtained previously for Eq.3.8.

The last component needed is K_d, which can be obtained from Eq.3.16 after extrapolating the needed coefficients (K_{di} and K_{do} from the corresponding plot in Roskam VI (*Fig.10.78b, Ch-10, p.485*).

$$K_d = \frac{K_{di}(1-\eta_i) - K_{do}(1-\eta_o)}{\eta_o - \eta_i} \tag{3.16}$$

where:

$$\begin{aligned}
K_d &= \text{Mean aerodynamic influence coefficient} \\
K_{di} &= \text{Inner aerodynamic influence coefficient} \\
K_{do} &= \text{Outer aerodynamic influence coefficient} \\
\eta_i &= \text{Aileron's Inner non-dimensional spanwise location} \\
\eta_o &= \text{Aileron's Outer non-dimensional spanwise location}
\end{aligned}$$

Finally, ΔC_{h_δ} can be obtained as well as the full estimation for C_{h_δ} can be obtained using Eq.3.10.

3.2 Tornado Implementation

The pre-existing Tornado MATLAB tool [10] was adapted to match the geometry definition used in the Roskam estimation, and the required inputs were extended as illustrated previously in Fig.2.7. This section presents the core foundations required to understand the program's structure and functioning.

To integrate the Tornado MATLAB code for the VLM into the previous implementation, several adjustments were necessary.
The main modifications introduced in this work can be summarized as follows:

- Integration with the Roskam-based input structure to ensure consistency between semi-empirical and numerical methods;
- Automated geometry and flight-state initialization, removing the dependency on the original text-based interface;
- Implementation of a dedicated hinge-moment extraction routine based on panel-wise force integration;
- Extension to support faster execution for lookup-table generation.

3.2.1 Geometry and State Definition

Firstly, the text-based user interface in the start-up menu of the program is updated to automatically register the current geometry data and flight state conditions, and the inputs required for the Roskam calculations are extended with the following variables illustrated in Tab.3.2.

— Flight State —		
M	Flight Mach number	–
Altitude	Geopotential altitude	m
V	True airspeed	m/s
α	Angle of attack	deg
β	Sideslip angle	deg
P	Roll rate	deg/s
Q	Pitch rate	deg/s
R	Yaw rate	deg/s

Table 3.2: Tornado flight state setup inputs

The previous requirements used in the Roskam estimation are integrated, while the deflection angle (δ) is separated as an input parameter for the state setup shown in Tab.3.2.

3.2.2 Lattice Generation

The functions used to perform the lattice calculations remain unchanged from the original Tornado code. The program provides the option to select different mesh distributions for each wing and partition defined (see Tab. 3.3). As mesh distribution is by default set as half-cosine, cosine respectively for spanwise and chordwise directions, but can also be set to the different options illustrated in Tab.3.3.

Mesh Distribution Type	Notes
Linear	Standard spacing
Spanwise half-cosine	Tip refinement
Spanwise half-cosine, chordwise cosine	Full edge refinement
Spanwise cosine	Symmetric refinement
Chordwise cosine	Leading-edge and trailing-edge refinement
3rd order center packing	Not suitable for wings

Table 3.3: Available mesh distribution options

Furthermore, two main methodologies can be selected to compute the lattice solution, whether the classical formulation of the Vortex Lattice method or the improved one, where the classical vortex arrangement has been replaced with a "vortex-sling" arrangement which basically works as the standard formulation, with the exception that the legs of the shoe are more flexible and consist of seven (instead of three) vortices of equal strength, as shown in Fig.3.6.

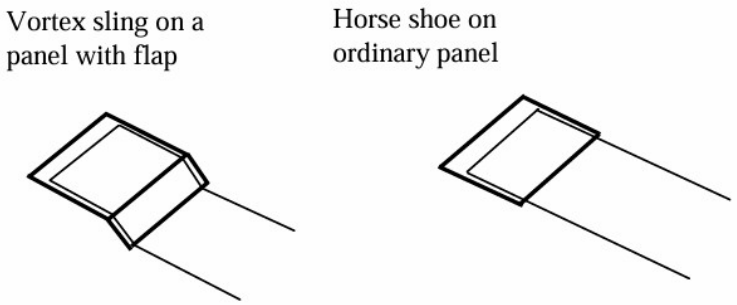


Figure 3.6: The seven-segment vortex sling vs the horse shoe[10]

3.2.3 Calculation of Control Surface Hinge Moments

The hinge moment is evaluated using a discrete formulation based on previously obtained data. In this approach, the aerodynamic force acting on each flap panel is obtained from the VLM solution, and the corresponding moment contribution is computed with respect to the geometrically defined hinge axis.

The free-stream dynamic pressure is defined as

$$q_\infty = \frac{1}{2}\rho V^2 \tag{3.17}$$

where ρ denotes the air density and V is the free-stream velocity.

For each control-surface panel, the force vector $\vec{F}_i$ is extracted from the aerodynamic solution. The representative pressure-center location is approximated by the midpoint of the bound vortex segment, i.e.

$$\vec{P}_{CP,i} = \frac{1}{2}\left(\vec{P}_{4,i} + \vec{P}_{5,i}\right) \tag{3.18}$$

where $\vec{P}_{4,i}$ and $\vec{P}_{5,i}$ denote the coordinates of the corresponding vortex endpoints.

The hinge axis is defined by two geometric points, $\vec{A}_h$ and $\vec{B}_h$, such that the unit vector along the hinge line is

$$\hat{h} = \frac{\vec{B}_h - \vec{A}_h}{\|\vec{B}_h - \vec{A}_h\|} \tag{3.19}$$

The orthogonal projection of $\vec{P}_{CP,i}$ onto the hinge line yields the pivot point

$$\vec{P}_{pivot,i} = \vec{A}_h + \left[(\vec{P}_{CP,i} - \vec{A}_h) \cdot \hat{h}\right] \hat{h} \tag{3.20}$$

and the corresponding moment arm is

$$\vec{r}_i = \vec{P}_{CP,i} - \vec{P}_{pivot,i} \tag{3.21}$$

The panel-wise hinge-moment contribution is then computed as

$$\vec{M}_i = \vec{r}_i \times \vec{F}_i \tag{3.22}$$

and the total hinge moment is obtained by summing the contributions from all flap panels and projecting the result onto the hinge axis:

$$M_h = \sum_{i=1}^{n} \left(\vec{M}_i \cdot \hat{h} \right) \tag{3.23}$$

For symmetric configurations, the same procedure is applied to the mirrored control surface, using the corresponding mirrored hinge axis and geometric points. The resulting hinge moment coefficient is defined as

$$C_h = \frac{M_h}{q_\infty S_f c_f} \tag{3.24}$$

where S_f is the control-surface area and c_f is the mean flap chord.

3.2.4 Iteration Over Flight States

To assemble a final lookup table, the full calculation process is repeated for a set of state conditions, obtaining a range of cases identified by a unique ID and representing different combinations of the inputs reported in Tab.3.2, with the only variation of Mach number, altitude, angle of attack, and aileron deflection.
The final output will then be constituted by:

$$N_{cases} = n_{Mach} * n_{altitudes} * n_{AoA} * n_{deflections} \tag{3.25}$$

3.3 CPACS

The Common Parametric Aircraft Configuration Schema (CPACS) is a standardized, XML-based data model designed to enable consistent, interoperable representation of aircraft configurations across multidisciplinary design environments. It provides a hierarchical and semantically structured framework in which all relevant aspects of an aircraft, geometry, systems, and mission definitions are described parametrically, making it particularly suitable for multidisciplinary design optimization (MDO) and collaborative development processes [12].

Within the proposed framework depicted in Fig. 3.7, CPACS serves as the common interface between the aircraft geometry definition and both estimation modules (Roskam and Tornado). Geometry is defined once in a single CPACS file and automatically extracted at runtime, eliminating the need to manually re-enter parameters across tools.



Figure 3.7: CPACS data extraction in MATLAB

3.3.1 File Structure and Data Sources

The implemented CPACS reader combines data from two complementary regions of the XML file, as illustrated in Fig. 3.8:

1. **Standard CPACS hierarchy** - Wing geometry, reference quantities, control-surface definitions and mass properties are read from their native locations within the `/cpacs/vehicles/aircraft/model/` and `/cpacs/vehicles/aircraft/model/wings/` branches.

2. **Tool-specific branch** - A small set of Tornado solver settings (panel counts, mesh type, and flap and symmetry flags) that lack a standard CPACS equivalent is stored under `/cpacs/toolspecific/tool/`. The design rationale and full parameter list are discussed in Section 3.3.3.

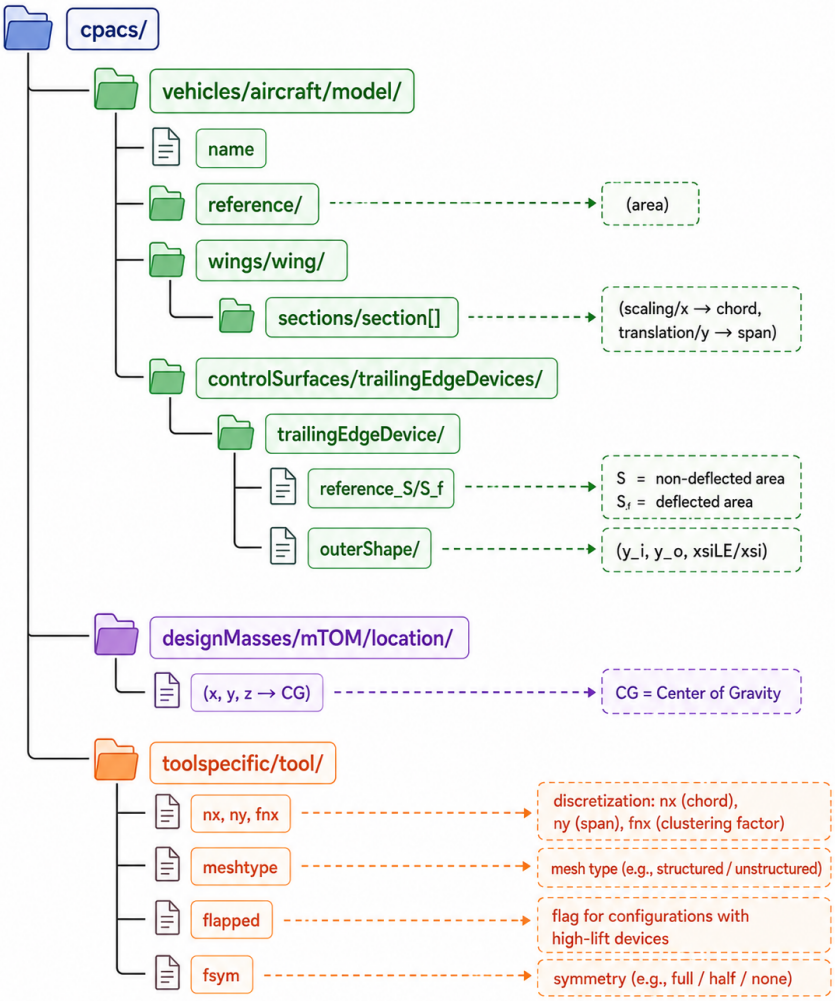


Figure 3.8: CPACS tree node hierarchy used by the extraction function

3.3.2 Automated Geometry Extraction

Geometric and aerodynamic parameters are extracted through the dedicated MATLAB function `readGeoFromCPACS_2`, which uses the path to trace back each variable in the XML document after the needed locations were manually traced and registered in an Excel file, as shown in Appendix 8.2. This approach allows each parameter to be retrieved by its full logical path, independently of file length or node ordering. The extraction proceeds in the following sequential stages, with the complete Path-to-field mapping summarized in Table 3.4.

Aircraft identification

The aircraft name is read from `/cpacs/vehicles/aircraft/model/name` and stored in the output structure for traceability across lookup-table entries.

Wing geometry

The half-span is derived from the `translation/y` component of the last wing section (`section[last()]`), and doubled to obtain the full span b. Root and tip chords are obtained from the corresponding `scaling/x` values of the first and last sections, respectively. The reference wing area S is read directly from `/cpacs/.../reference/area`. The taper ratio is then computed internally as $\lambda = c_t / c_r$.

Aileron geometry

The aileron surface area S_f is read from the `trailingEdgeDevice/reference_S/S_f` node. Spanwise boundary coordinates y_i and y_o are retrieved from the `outerShape/innerBorder` and `outerBorder` nodes, respectively, and the aileron span is computed as $b_f = y_o - y_i$. The chord fraction $\bar{c}_f/c$ is derived from the `xsiLE/xsi` parameter as $\bar{c}_f/c = 1 - \xi_{\text{LE}}$.

Airfoil UIDs

Root and tip airfoil identifiers are read from the `airfoilUID` nodes of the first and last wing sections, respectively, and stored as string fields (`Geo.Airfoil_root`, `Geo.Airfoil_tip`). These UIDs are passed directly to the Tornado geometry builder to retrieve the corresponding profile coordinates from the CPACS airfoil database.

Center of gravity

The CG position (x, y, z) is read from `designMasses/mTOM/location`, providing the reference point for moment calculations in Tornado.

Wing partition decomposition

To interface with the Tornado VLM, which requires an explicit partition-by-partition description of the wing, the function decomposes the half-span into two segments: an inboard clean partition (from root to y_i) and an outboard flapped partition (from y_i to y_o). Leading-edge offsets, spans, taper ratios, sweep angles, and dihedral angles are computed for each partition and stored as arrays within the `Geo` structure.

Tornado solver settings

Panel resolution parameters and mesh flags are read from the `toolspecific` branch and stored in the `Geo` structure, allowing the solver to be configured entirely from the CPACS file. The rationale for storing these parameters outside the standard hierarchy, together with the full parameter list, is discussed in Section 3.3.3.

Table 3.4: Mapping between CPACS Path and `Geo` structure fields.

CPACS Path	Geo struct field
`.../reference/area`	`Geo.S`
`.../section[last()]/transformation/translation/y * 2`	`Geo.b`
`.../section[1]/transformation/scaling/x`	`Geo.c_r`
`.../section[last()]/transformation/scaling/x`	`Geo.c_t`
`.../section[1]/airfoilUID`	`Geo.Airfoil_root`
`.../section[last()]/airfoilUID`	`Geo.Airfoil_tip`
`.../trailingEdgeDevice/reference_S/S_f`	`Geo.S_f`
`.../innerBorder/y_i`	`Geo.y_i`
`.../outerBorder/y_o`	`Geo.y_o`
`.../innerBorder/xsiLE/xsi`	`Geo.fc` $(= 1 - \xi_{\text{LE}})$
`.../mTOM/location/{x,y,z}`	`Geo.{x,y,z}_CG`
`toolspecific/tool/nx`	`Geo.nx`
`toolspecific/tool/ny`	`Geo.ny`
`toolspecific/tool/fnx`	`Geo.fnx`
`toolspecific/tool/meshtype`	`Geo.meshtype`
`toolspecific/tool/flapped`	`Geo.flapped`
`toolspecific/tool/fsym`	`Geo.fsym`

3.3.3 Tool-Specific Extension

A limited set of parameters required by the Tornado solver, panel counts, mesh distribution type, and symmetry and flap flags, has no equivalent in the standard CPACS schema. These are stored in a dedicated `<tool>` node under the `toolspecific` branch, as shown in Table 3.5. This design keeps the solver fully configurable from the CPACS file while containing the non-standard content within the branch that CPACS reserves precisely for this purpose.

Table 3.5: Tool-specific CPACS parameters for the Tornado solver.

CPACS node	Geo field	Description
`nx`	`Geo.nx`	Chordwise panel count
`ny`	`Geo.ny`	Spanwise panel count
`fnx`	`Geo.fnx`	Flap chordwise panels
`meshtype`	`Geo.meshtype`	Distribution type index
`flapped`	`Geo.flapped`	Flap-active flag (boolean)
`fsym`	`Geo.fsym`	Symmetry flag (boolean)

In future developments, the `toolspecific` content can be progressively reduced as CPACS libraries are integrated, allowing panel counts and mesh settings to be derived automatically from the native geometry rather than declared explicitly.

3.3.4 Output Structure

The complete output of `readGeoFromCPACS_2` is a single MATLAB structure `Geo` containing all parameters required by both the Roskam and Tornado modules. Its fields are grouped by category in Table 3.6, together with the module that consumes each group.

Table 3.6: Output fields of the `Geo` structure populated from CPACS.

Category	Fields	Consumer
Wing planform	`b, S, c_r, c_t, TR_full, Lambda`	Roskam, Tornado
Aileron geometry	`S_f, y_i, y_o, bf, fc, hinge`	Roskam, Tornado
CG / reference	`x_CG, y_CG, z_CG, x/y/z_ref`	Tornado
Wing partitions	`startx/y/z, span, TR, Lambda_c4, Gamma`	Tornado
Airfoil UIDs	`Airfoil_root, Airfoil_tip, Airfoil`	Tornado
Solver settings	`nx, ny, fnx, meshtype, flapped, fsym`	Tornado

Once populated, the `Geo` structure is passed directly into both estimation methods without further adjustments, ensuring that both methods operate on an identical, consistently sourced geometric description throughout the lookup-table generation loop.

4 Results and Comparison

This chapter presents the results of the comparison study of the *F/A-18 Systems Research Aircraft (SRA)*, using experimental flight-test data reported by Navarro [13] as the primary benchmark. The comparative study aims to assess the consistency and accuracy of the two hinge-moment estimation approaches developed in this thesis: the automated Roskam semi-empirical method and the Tornado vortex-lattice panel method.

The chapter is organized as follows. Section 4.1 details the modeling assumptions for each method. Section 4.2 defines the reference geometry, panel mesh, and the four flight conditions examined. Section 4.3 presents the core comparison results for the two subsonic conditions, with comparisons to both the NASA flight data and the reference simulation of Chakraborty et al. [11]. Section 4.4 addresses the two supersonic conditions, for which no simulation reference is available, and shifts the focus to assessing method behavior beyond the linear validity range. Section 4.5 complements the time-domain comparison with supporting aerodynamic analyses: non-dimensional hinge-moment coefficients and scatter correlation with the flight data. Finally, Section 4.6 presents two extended parametric sweeps over a broader Mach envelope, providing lookup-table heatmaps across the full subsonic-to-supersonic range and characterizing the divergence between the two methods in the absence of flight data.

4.1 Assumptions

The comparison presented in this chapter should be interpreted as a method comparison for a single reference configuration, not as a general proof of accuracy across fighter aircraft. The chosen test case provides a relevant benchmark, but broader conclusions would require additional geometries, more mesh-sensitivity analysis, and independent comparison cases.

Both methodologies share a common set of standard assumptions related to inviscid, attached-flow aerodynamics and linear control-surface behavior. These are generally acceptable at low-to-moderate angles of attack and subsonic speeds, but may not hold in the high-speed, high-α regimes typical of fighter operations. In addition to these shared assumptions, each method introduces its own method-specific simplifications, described in the following subsections.

4.1.1 Roskam Method

The Roskam methodology was integrated into the automated framework under the following assumptions:

- **Round leading-edge nose type**: the most common configuration for conventional ailerons, it was assumed.
- **Digitized plot accuracy**: the empirical charts from Roskam Volume VI [4], reproduced digitally in MATLAB as shown in Fig. 3.3, was implemented with a level of resolution appropriate for conceptual design. Higher-fidelity digitization could be introduced in future work to reduce interpolation errors.
- **Trailing-edge angle approximation**: the Φ_{TE} condition in Eq. 3.5 is verified numerically by checking whether the mean of the two angle approximations satisfies the condition within a prescribed tolerance. Depending on the airfoil coordinate density, this step may introduce small approximation errors, particularly for non-standard profiles.
- **Linear aerodynamic regime**: the method assumes a linear relationship between the hinge-moment coefficient, the angle of attack, and the control-surface deflection. Accuracy degrades for angles of attack above approximately 12° or deflections beyond approximately 20°, where flow separation and nonlinear boundary-layer effects become significant.
- **Static evaluation**: Roskam coefficients $C_{h\alpha}$ and $C_{h\delta}$ are computed once for each flight condition and then applied linearly. The method does not account for dynamic changes in angle of attack or deflection history within a manoeuvre.
- **Prandtl-Glauert compressibility correction**: the method applies a $1/\sqrt{1-M^2}$ correction factor to account for subsonic compressibility effects. This correction becomes imaginary for $M > 1$ and is therefore physically invalid in the supersonic regime; results for cases (c) and (d) must be interpreted accordingly.

4.1.2 Tornado Panel Method

The Tornado VLM implementation was configured as described in Section 3.2, with the following fixed assumptions:

- **Inviscid potential flow**: the solver neglects viscous effects, boundary-layer development, and flow separation, which limits its applicability to attached-flow conditions.
- **Flexible wake model**: Tornado's flexible wake option was used, allowing the trailing wake geometry to vary with flight conditions. This improves the accuracy of wing–stabiliser interaction predictions compared to a rigid-wake formulation.
- **Mesh configuration**: a half-cosine spanwise and cosine chordwise mesh distribution was adopted, providing refinement near the wing tip and at the leading and trailing edges. The sensitivity of the hinge moment predictions to mesh refinement was not systematically investigated in this work and represents a topic for future study.
- **Symmetric configuration**: the hinge moment is computed on a single aileron panel and mirrored symmetrically, assuming no asymmetric aerodynamic loading.
- **Static angle of attack**: as the Tornado solver computes a steady potential-flow solution, the angle of attack is held constant for each evaluation point. A representative fixed angle of attack is selected through the sensitivity analysis described in Section 4.3.1.

- **Prandtl-Glauert compressibility correction**: this method applies as well by default a $1/\sqrt{1-M^2}$ correction factor to account for subsonic compressibility effects. This correction becomes imaginary for $M > 1$ and is therefore physically invalid in the supersonic regime; results for cases (c) and (d) must be interpreted accordingly.
- **Fixed TR and Sweep**: this added assumption is based on the code functioning and might be expanded in future studies, taking the CPACS convention more standardized section division into account.

4.2 Test Case Definition

The reference aircraft is the *F/A-18 SRA*, for which Chakraborty et al. [11] provide hinge moment predictions from a high-fidelity modeling and simulation environment utilizing flight mechanics (6 DoF) and actuator models, and Navarro [13] reports experimental actuator performance data from NASA flight tests.

4.2.1 Geometry

The wing and aileron geometry used as input for both methods is summarized in Table 4.1. The values were derived from publicly available F/A-18 geometric data and are consistent with the parameters used in the reference study [11].

Table 4.1: F/A-18 SRA geometry inputs used for hinge moment estimation, obtained from [13] and from publicly available F/A-18 Hornet specifications published by the Federation of American Scientists (FAS) [14].

Parameter	Description	Value	Unit
b	Wing span	12.31	m
S	Wing reference area	37.16	m^2
c_r	Root chord	4.04	m
c_t	Tip chord	1.68	m
$\Lambda_{c/4}$	Quarter-chord sweep angle	20	deg
Γ	Wing dihedral angle	-3	deg
y_i	Inboard aileron spanwise location	4.03	m
y_o	Outboard aileron spanwise location	5.87	m
S_f	Aileron surface area	1.135	m^2
c_f/c	Aileron-to-wing chord ratio	0.288	–
Hinge position	Hinge location (fraction of chord)	25	%c

The wing planform definition and aileron partition used in Tornado implementation are illustrated in Fig. 4.1. The figure shows the top view of the wing, including the inboard and outboard spanwise limits of the aileron (y_i and y_o), the hinge line position, and the chord distribution from root to tip. Given the assumption of fixed taper ratio and sweep, it follows that, at the current state, Tornado assumes that the only two sections will be constituted by the flapped and non-flapped parts of the half span, as shown in Fig. 4.1.

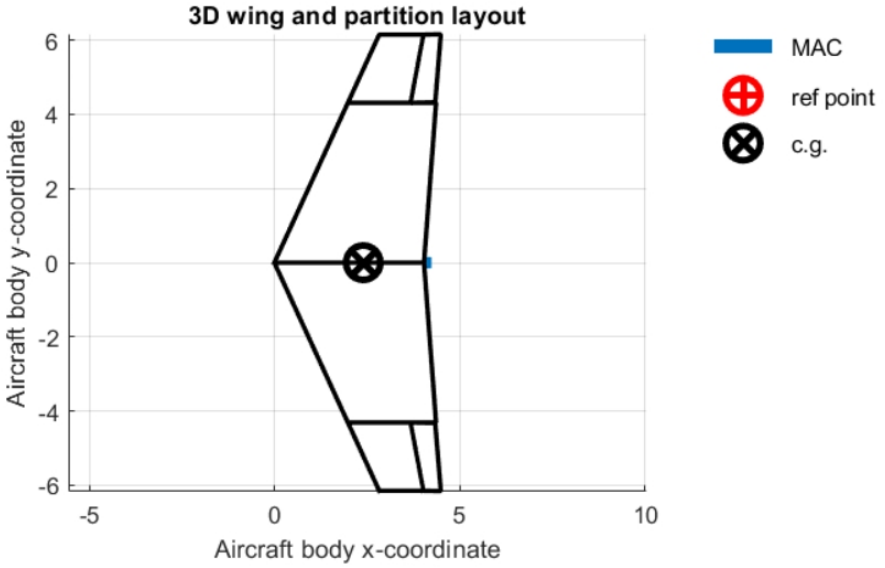


Figure 4.1: F/A-18 SRA wing top view and aileron partition definition used as input for both the Roskam and Tornado estimation methods.

4.2.2 Tornado Mesh

The Tornado vortex lattice mesh was generated from the geometry of Table 4.1 and is shown in Fig. 4.2. The panel layout illustrates the spanwise and chordwise discretization adopted for the wing and the aileron partition. The aileron panels, identified by the deflected trailing segment, are highlighted to indicate the region over which the hinge moment integration described in Section 3.2.3 is performed.

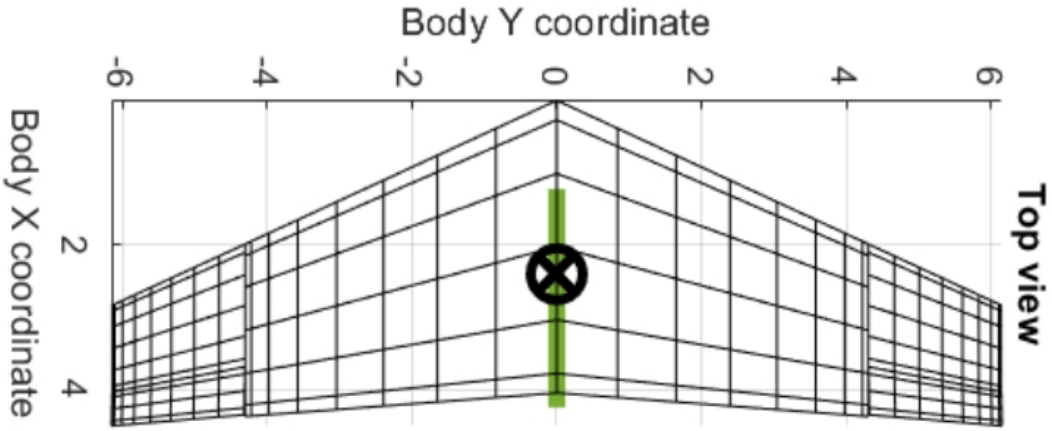


Figure 4.2: Tornado vortex lattice mesh for the F/A-18 SRA wing geometry. Mesh distribution: half-cosine spanwise, cosine chordwise.

4.2.3 Flight Conditions

Four flight conditions were selected to match those reported in [11, 13], enabling a direct comparison between the methods developed in this thesis, the reference simulation results (conditions (a)-(b) only), and the NASA flight test data (all four conditions). The conditions are summarized in Table 4.2.

Table 4.2: Flight conditions used for the comparison case study.

ID	Condition	Mach	Altitude [kft]	AoA [deg]	Deflection [deg]
(a)	Abrupt roll doublet	0.72	25	$[-5, +5]$	$[-20, +20]$
(b)	Abrupt aileron reversal (full stick)	0.84	25	$[-5, +5]$	$[-20, +20]$
(c)	Aileron reversal (half stick)	1.58	34	$[-5, +5]$	$[-15, +15]$
(d)	Abrupt aileron reversal (full stick, right)	1.60	34	$[-5, +5]$	$[-15, +15]$

Cases (a) and (b) are subsonic and fall within the linear validity range of both methods; they constitute the primary comparison benchmark. Cases (c) and (d) are supersonic: the Prandtl-Glauert correction breaks down for $M > 1$, both for Roskam and Tornado, which already collapses with the hypothesis of potential flow theory breaking. These two conditions, therefore, serve to characterize the limitations of each method rather than to validate them quantitatively.

All four maneuvers involve rapid, large-amplitude aileron deflections with simultaneously varying angle of attack. The left aileron deflection time histories for all four maneuvers are shown in Fig. 4.3.

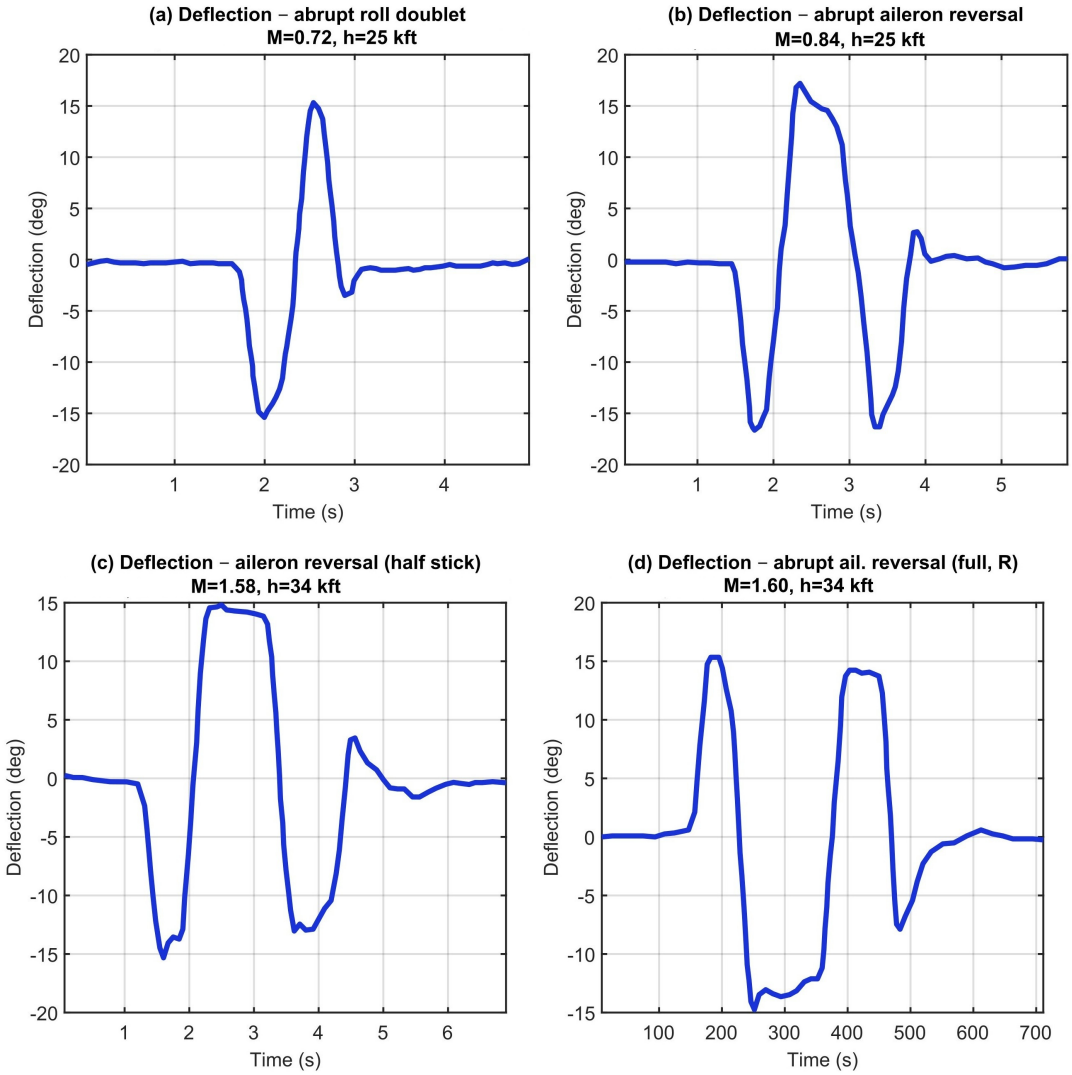


Figure 4.3: Left aileron deflection over time for the four maneuvers considered. Top row: cases (a) and (b) at 25 kft. Bottom row: cases (c) and (d) at 34 kft.

Because the Roskam and Tornado methods provide static, steady-state estimates, the dynamic hinge moment is reconstructed by interpolating the pre-computed lookup table as a function of the instantaneous deflection angle at each time step, under a fixed representative angle of attack. The selection procedure for this angle is described in Section 4.3.1.

4.3 Comparison: Subsonic Flight Conditions

This section presents the primary comparison results for cases (a) and (b), the two subsonic conditions for which the reference simulation of Chakraborty et al. [11] is available alongside the NASA flight data.

4.3.1 Angle-of-Attack Sensitivity and Best-Alpha Selection

Since the flight-test dataset does not provide time-dependent angle-of-attack measurements, a direct semi-steady reconstruction of the hinge moment history is not possible with a varying α. Instead, an equivalent static angle of attack is identified for each maneuver by minimizing the RMS error between the NASA-measured hinge moment and the Tornado-based reconstruction.

The root-mean-square (RMS) error and the mean signed error between the Tornado hinge moment reconstruction and the NASA flight data [13] were computed for each angle of attack available in the lookup table. The results are shown in Fig. 4.4.

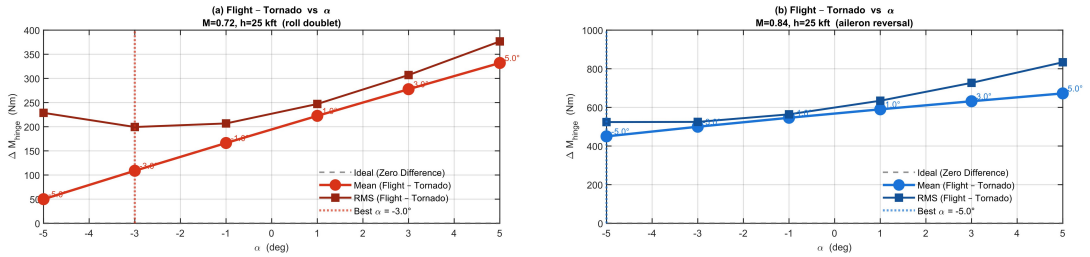


Figure 4.4: RMS and mean errors between the Tornado hinge moment reconstruction and the NASA flight data, as a function of the assumed angle of attack. Left: Case (a), Mach 0.72, roll doublet. Right: Case (b), Mach 0.84, aileron reversal. The dashed vertical line marks the angle of attack selected for the main comparison.

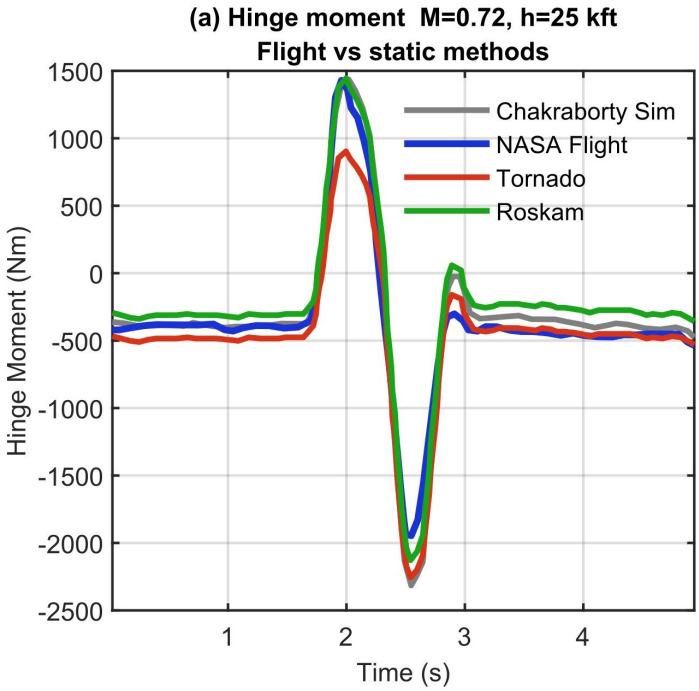
The RMS error curves exhibit a clear minimum for both cases. For case (a) at Mach 0.72, the minimum RMS error of 199 Nm is achieved at $\alpha = -3^\circ$. For case (b) at Mach 0.84, the minimum RMS error of 524 Nm corresponds to $\alpha = -5^\circ$. These angles from a physical point of view might seem unrealistic for the maneuvers considered, confirming that the approach used only serves as an error minimizer to show optimistic estimations, and further developments could benefit this angle derivation.

The selected angle of attack was used to identify a representative operating point that minimizes discrepancy with the available reference data. This choice is intended to compare model behavior under comparable conditions, although it does not constitute an independent calibration and therefore should not be interpreted as a fully blind comparison.

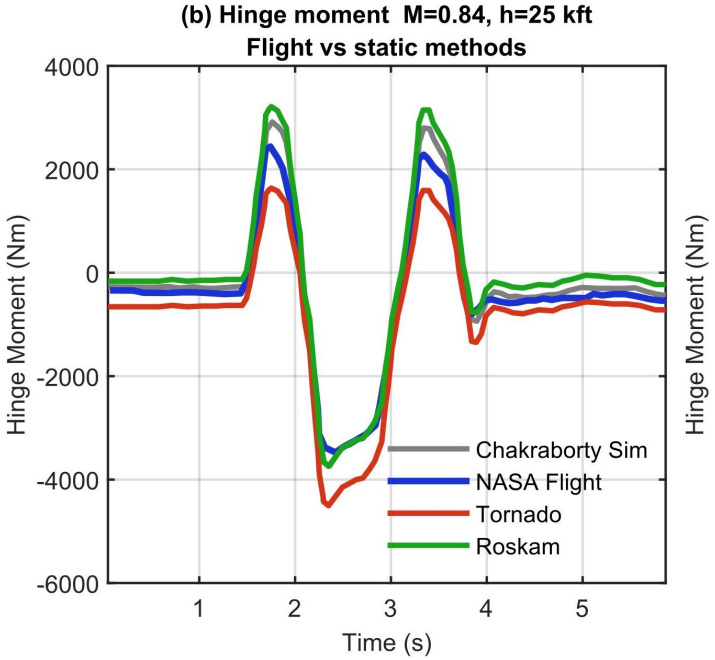
4.3.2 Dynamic Hinge Moment Comparison

Using the angles of attack identified in Section 4.3.1, the hinge moment time history is reconstructed for both methods by interpolating the lookup table at the instantaneous aileron

deflection measured during each maneuver. Figure 4.5 presents the complete four-panel comparison for cases (a) and (b).



(a) Case (a) (Mach 0.72, roll doublet). Upper panel shows the aileron deflection input. Lower panel compares the NASA flight data [13], the reference simulation [11], the Tornado estimate, and the Roskam estimate, at the best-fit angle of attack identified in Fig. 4.4.



(b) Case (b) (Mach 0.84, aileron reversal). Upper panel shows the aileron deflection input. Lower panel compares the NASA flight data [13], the reference simulation [11], the Tornado estimate, and the Roskam estimate, at the best-fit angle of attack identified in Fig. 4.4.

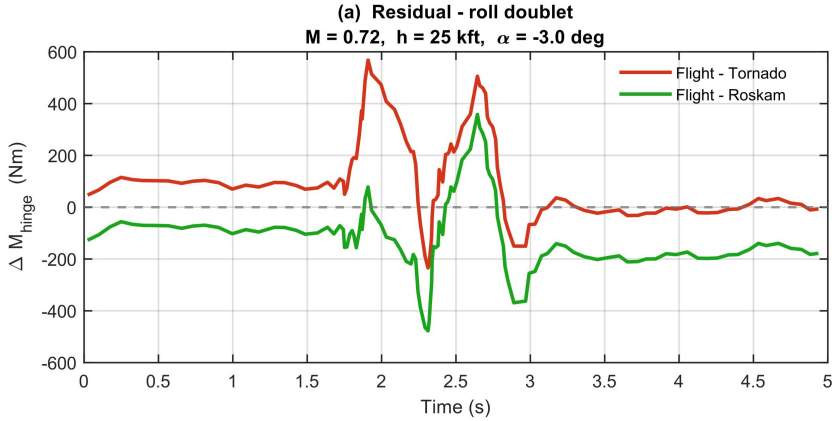
Figure 4.5: Hinge moment estimation comparison for Case (a) (Mach 0.72, roll doublet) and Case (b) (Mach 0.84, aileron reversal).

Case (a) - Roll doublet, Mach 0.72. The Roskam methodology remains fairly consistent throughout the full maneuver. Tornado underestimates the loads at positive peak deflections, with the discrepancy reducing at small deflections and during transition phases. Both methods perform comparably to the reference simulation by Chakraborty et al. [11] over most of the time window.

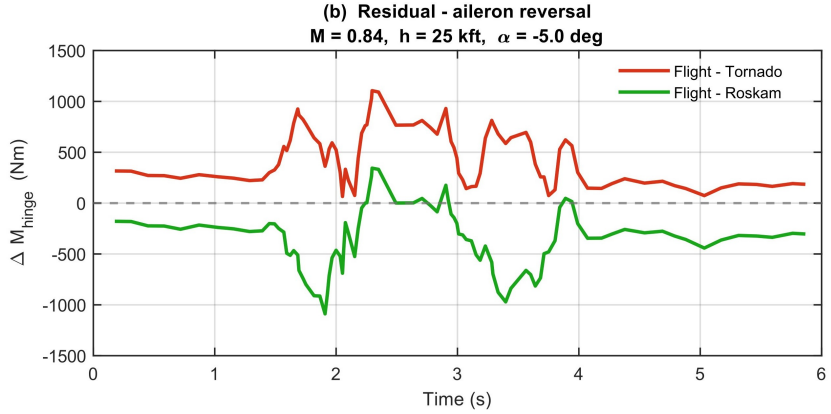
Case (b) - Aileron reversal, Mach 0.84. In the second case, both methods show a degradation in accuracy relative to case (a). The Roskam prediction deteriorates during peak positive deflections. Tornado, while broadly following the general trend, exhibits an enhanced systematic offset at both zero-deflection positions and at the peaks.

4.3.3 Residual Time History

To characterize the temporal distribution of the prediction error, the residual hinge moment, defined as the difference between the NASA flight measurement and each method's estimate, is plotted as a function of time in Fig. 4.6. This representation complements the aggregate RMS metrics by identifying whether the error is uniformly distributed across the maneuver or concentrated at specific instants, such as peak deflections or direction reversals.



(a) Case (a) (Mach 0.72, roll doublet). Residual hinge moment (NASA flight data minus method estimate) as a function of time, at the best-fit angle of attack identified in Fig. 4.4. The dashed horizontal line indicates zero residual.



(b) Case (b) (Mach 0.84, aileron reversal). Residual hinge moment (NASA flight data minus method estimate) as a function of time, at the best-fit angle of attack identified in Fig. 4.4. The dashed horizontal line indicates zero residual.

Figure 4.6: Residual hinge moment as a function of time for Case (a) (Mach 0.72, roll doublet) and Case (b) (Mach 0.84, aileron reversal).

4.3.4 Quantitative Error Summary - Subsonic Cases

Table 4.3 summarizes the RMS error and mean absolute error (MAE) for each method for the two subsonic cases, computed over the full duration of the respective maneuver time window.

Table 4.3: RMS, MAE and Max error summary for subsonic cases (a) and (b). All values in Nm.

	Tornado (best α)			Roskam (best α)			Chakraborty [11]		
Case	**RMS**	**MAE**	**Max**	**RMS**	**MAE**	**Max**	**RMS**	**MAE**	**Max**
(a) Mach 0.72	199.40	142.44	567.77	184.27	157.32	477.13	178.70	127.97	-
(b) Mach 0.84	524.05	450.05	1105.39	459.48	383.03	1088.55	279.99	214.68	-

4.4 Supersonic Cases: Method Limitations

Supersonic cases are mainly included to demonstrate how the methods behave beyond their intended range, not to assert physically accurate predictions in that regime. Since both Roskam's empirical correction and the VLM-based approach rely on assumptions invalid for $M > 1$, their results should be seen as boundary-case examples of method failure rather than confirmed aerodynamic data.

Cases (c) and (d) are flown at Mach 1.58 and Mach 1.60, respectively, at an altitude of 34 kft. No reference simulation is available for these conditions. The NASA flight data serve as the sole reference. Both conditions exceed the physical validity range of both methods: the Roskam Prandtl-Glauert correction yields complex values for $M > 1$, and the Tornado VLM is inherently an incompressible solver that produces a Mach-independent result. These cases are therefore not treated as quantitative comparison benchmarks but rather as illustrations of method behavior outside the design envelope.

4.4.1 Dynamic Comparison - Supersonic Cases

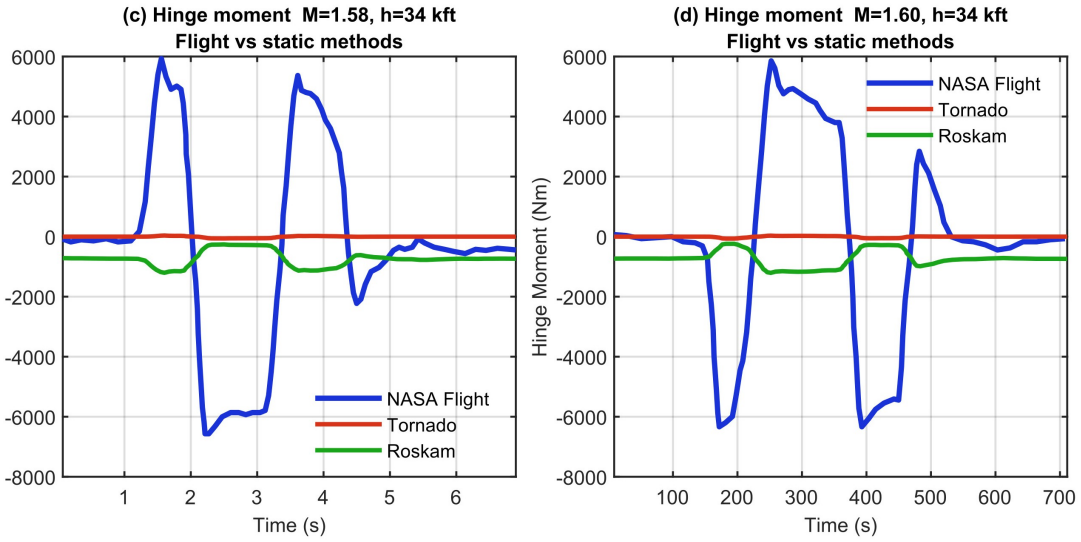


Figure 4.7: Hinge moment estimation comparison for Case (c) (Mach 1.58, aileron reversal, half stick, left) and Case (d) (Mach 1.60, abrupt aileron reversal, full stick right, right). No simulation reference is available. Roskam values shown are $\mathrm{Re}[C_h]$ where the Prandtl–Glauert correction yields complex numbers.

Case (c) - Aileron reversal (half stick), Mach 1.58. In the first supersonic case, both static methods fail to capture the hinge moment magnitude observed in the NASA flight data. Tornado predicts values close to zero throughout the entire maneuver, showing no sensitivity to the aileron inputs. Roskam produces a small negative offset of approximately -500 to -1000 Nm, which remains nearly constant and does not reproduce the dynamic peaks of ± 6000 Nm present in the flight record.

Case (d) - Aileron reversal (full stick, right), Mach 1.60. The second supersonic case confirms the trends observed in case (c). Tornado remains essentially flat at zero across the full time window, while Roskam reproduces only a small quasi-steady negative offset. Neither method captures the large oscillatory peaks present in the NASA flight data, which reach magnitudes of approximately ± 6000 Nm. The flight data in this case also exhibits a significantly longer time scale compared to case (c), with the maneuver extending over several hundred seconds.

4.4.2 Quantitative Error Summary - Supersonic Cases

As expected from the simulation results the RMS, mean and maximum errors are reported in Tab.4.4 for cases (c)-(d), confirming significant lack of accuracy or complete absence of sensitivity to the inputs registered.

Table 4.4: RMS, MAE, and Max error summary for supersonic cases (c) and (d). All values in Nm.

	Tornado (best α)			Roskam (best α)		
Case	**RMS**	**MAE**	**Max**	**RMS**	**MAE**	**Max**
(c) Mach 1.58	3322.13	2554.86	6532.25	3600.83	2739.42	6918.81
(d) Mach 1.60	3339.21	2572.49	6298.41	3635.95	2804.13	6674.12

4.5 Additional Aerodynamic Analysis

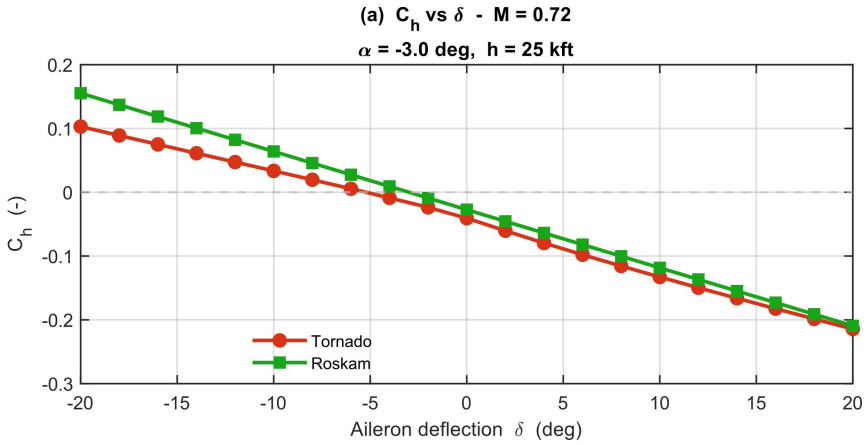
4.5.1 Non-Dimensional Hinge Moment Coefficient

The dimensional hinge moment M_h is scaled by the dynamic pressure q_∞, the aileron area S_f, and the mean flap chord c_f to obtain the non-dimensional coefficient:

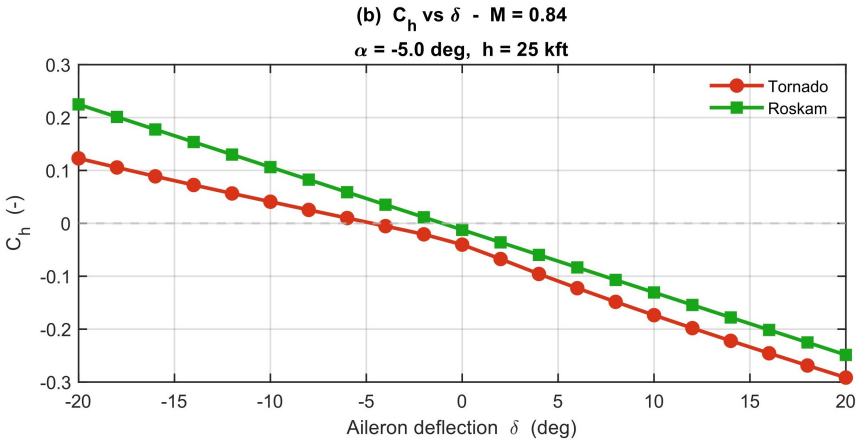
$$C_h = \frac{M_h}{q_\infty S_f c_f} \tag{4.1}$$

This quantity isolates the aerodynamic contribution from the flight-condition-dependent scaling, enabling a direct comparison of the two models independent of Mach number and altitude effects.

Figure 4.8 shows C_h as a function of aileron deflection for both subsonic cases, at the respective best-fit angle of attack. Supersonic cases are not included due to the high discrepancy between the methods, due to the method breakdown previously presented in Fig.4.7.



(a) Case (Mach 0.72, roll doublet). Non-dimensional hinge moment coefficient C_h as a function of aileron deflection δ, at the respective best-fit angle of attack. Cross markers indicate complex Roskam values due to Prandtl–Glauert breakdown ($M > 1$).



(b) Case (Mach 0.84, aileron reversal). Non-dimensional hinge moment coefficient C_h as a function of aileron deflection δ, at the respective best-fit angle of attack. Cross markers indicate complex Roskam values due to Prandtl–Glauert breakdown ($M > 1$).

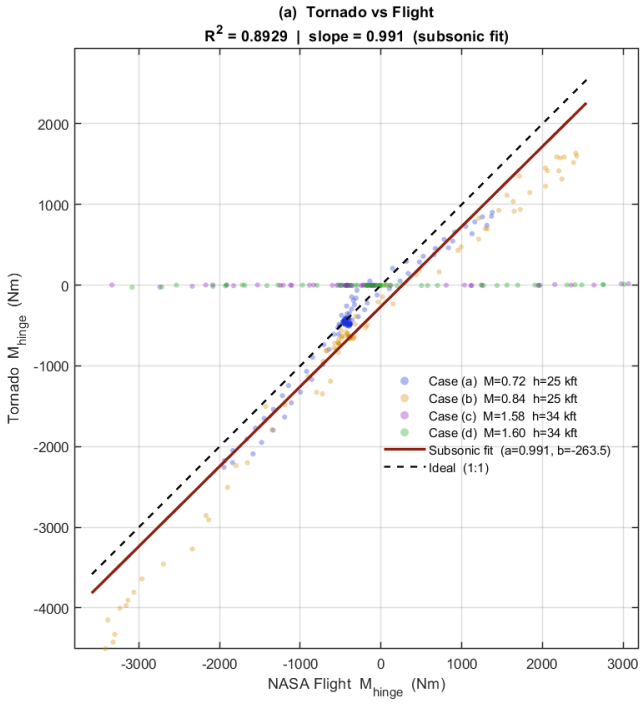
Figure 4.8: Non-dimensional hinge moment coefficient C_h as a function of aileron deflection δ for Case (a) (Mach 0.72, roll doublet) and Case (b) (Mach 0.84, aileron reversal).

Cases (a) and (b) - Subsonic regime. In both subsonic cases, Tornado and Roskam predict a negative and approximately linear variation of C_h with deflection, consistent with the expected sign of the hinge moment derivative C_{h_δ}. The two methods yield similar slopes, with Roskam producing a systematically higher offset at negative deflections.

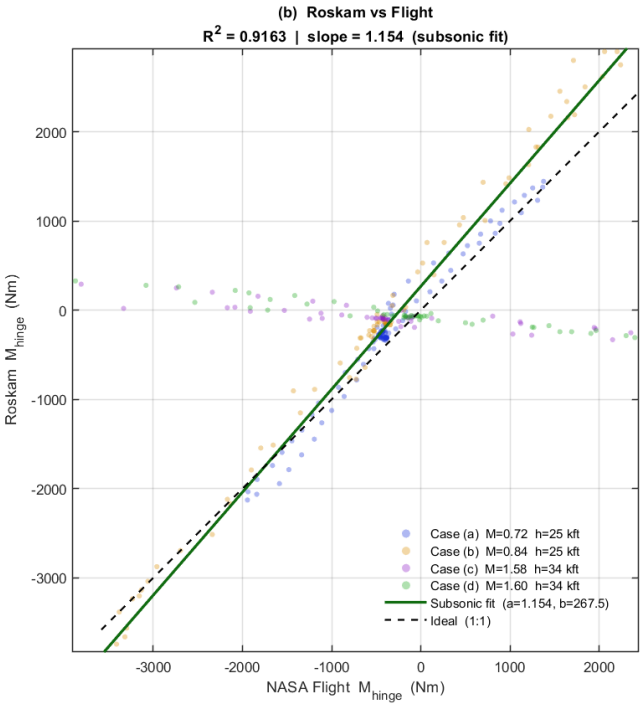
Supersonic regime. In the supersonic cases, the overall magnitude of C_h is substantially oscillating around zero, since the completely off predictions result in a reduced coefficient compared to the subsonic regime.

4.5.2 Scatter Correlation Analysis

Figure 4.9 presents scatter plots of the estimated hinge moment against the NASA flight measurement for each time step of all four maneuvers combined, color-coded by case. A perfect estimator places all points on the 45° diagonal. Deviations indicate systematic bias (offset) or random scatter (spread). Linear regression coefficients and the coefficient of determination R^2 are annotated as quantitative summary metrics.



(a) Tornado estimate. Scatter plot of estimated versus measured hinge moment, combining all time steps from all four cases. Each case is color-coded. The dashed line is the ideal 1:1 relationship; the solid line is the subsonic-case linear regression fit.



(b) Roskam estimate. Scatter plot of estimated versus measured hinge moment, combining all time steps from all four cases. Each case is color-coded. The dashed line is the ideal 1:1 relationship; the solid line is the subsonic-case linear regression fit.

Figure 4.9: Scatter plot of estimated versus measured hinge moment for Tornado and Roskam, combining all time steps from all four cases.

For Tornado, the subsonic cases (a) and (b) produce a tight clustering along the diagonal, indicating a close tracking of the flight loads. The regression restricted to these two cases yields a slope of 0.991, essentially coincident with the ideal unity, and a coefficient of determination of $R^2 = 0.8929$; the near-unit slope shows that the method reproduces the subsonic hinge-load magnitude with no significant scale bias, the local underprediction at the peak deflections visible in Fig. 4.5 averaging out over the maneuver. The supersonic cases (c) and (d), plotted for completeness but excluded from the fit, collapse to zero on the vertical axis regardless of the measured value, confirming the method's inability to reproduce the flight loads under those conditions.

Roskam shows a qualitatively similar clustering for the subsonic cases, though with greater scatter, with a subsonic-fit slope of 1.154 and $R^2 = 0.9163$. The slope above unity indicates a systematic overestimation of the hinge-load magnitude of roughly 15%, consistent with the higher offset already noted in the non-dimensional comparison of Fig. 4.8. Its supersonic points are again concentrated near zero on the estimated axis.

Both subsonic fits therefore lie close to the ideal 1:1 line, in line with the comparable RMS errors reported in Table 4.3. The regression is intentionally restricted to the subsonic regime: including the supersonic cases would bias the fit toward a near-zero slope and a non-informative R^2 dominated by the failure mode rather than by the subsonic performance the metric is meant to characterize. In both panels, the deviation of the supersonic points from the ideal 1:1 line remains the most evident feature, as they yield the largest measured values and are systematically driven to zero by both methods.

4.6 Extended Mach Envelope Parametric Study

The comparison study of Section 4.3 used a high-resolution parameter sweep ($\approx$1000 cases) focused on the four specific Mach-Altitude combinations from the NASA dataset. To characterize how the two methods behave across the full subsonic-to-supersonic transition, two complementary parametric sweeps were performed over a broader Mach range with a reduced grid of deflection and angle-of-attack samples. These sweeps do not have corresponding flight data for comparison; their purpose is to characterize the methodological behavior and mutual divergence as a function of Mach number.

4.6.1 Simulation Setup

Two extended parameter sweeps were conducted to investigate the behavior of the estimation methods beyond the four validated flight conditions. The first covers the subsonic regime with increased Mach resolution around the two validated subsonic points; the second bridges the transonic gap and extends through the supersonic regime up to the two validated supersonic conditions. Both sweeps share the same altitude levels and use a coarser angle-of-attack and deflection grid than the primary comparison run to limit computational cost.

The parameter combinations for each sweep are summarized in Table 4.5. Each configuration consists of all combinations of the listed values, yielding approximately 486 cases for the subsonic sweep and 462 cases for the transonic–supersonic sweep, with an estimated runtime of approximately 15 minutes per sweep.

Table 4.5: Parameter grids for the two extended simulation sweeps.

Sweep	Mach values	Alt. [kft]	AoA [deg]	Cases
Subsonic	0.50, 0.60, 0.65, 0.70, 0.72, 0.76, 0.80, 0.84, 0.88	25, 34	$-5,\ 0,\ +5$	≈ 486
Trans./Sup.	0.90, 0.95, 1.00, 1.05, 1.10, 1.20, 1.30, 1.40, 1.50, 1.58, 1.60	25, 34	$-5,\ 0,\ +5$	≈ 462

Deflection grid: -20 to $+20$ in steps of 5 (subsonic); -15 to $+15$ in steps of 5 (transonic/supersonic).

The subsonic sweep includes the two validated Mach numbers ($M = 0.72$ and $M = 0.84$) as interior points of the grid, so that the heatmap produced from this sweep can be directly compared against the comparison results of Section 4.3. Similarly, the transonic–supersonic sweep is anchored at $M = 1.58$ and $M = 1.60$, which correspond to the validated supersonic conditions and serve as reference points for the extended heatmap.

4.6.2 Lookup Table Heatmaps

Figures 4.10 and 4.11 visualize the hinge moment coefficient lookup table as a two-dimensional color map over the Mach number and aileron deflection parameter space, at fixed angle of attack, for the subsonic and transonic-supersonic sweeps respectively. The boundary between the two sweeps falls in the range $0.88 < M < 0.9$. Dashed white lines mark the four validated Mach numbers.

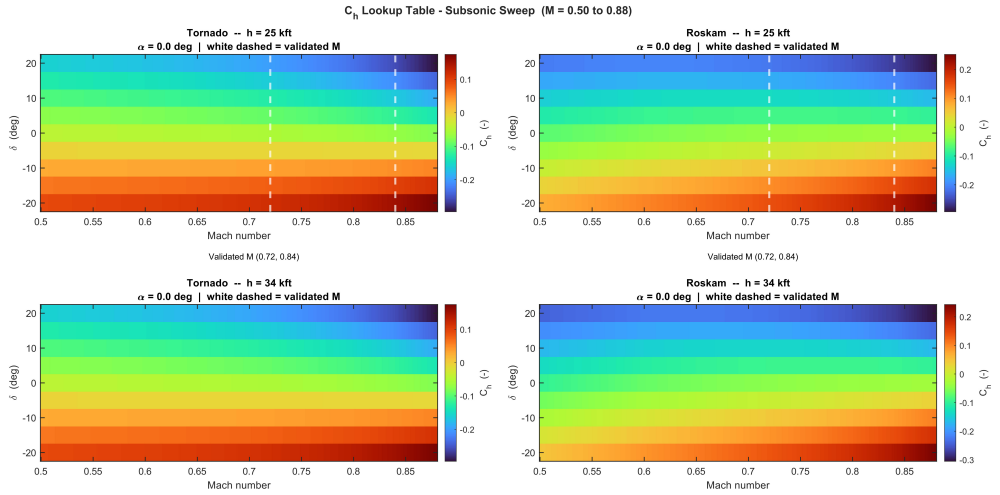


Figure 4.10: Hinge moment coefficient C_h as a function of Mach number and aileron deflection - subsonic sweep ($M = 0.50$–0.88). Rows: altitude (25 kft top, 34 kft bottom). Columns: method (Tornado left, Roskam right). White dashed lines mark the two validated subsonic Mach numbers.

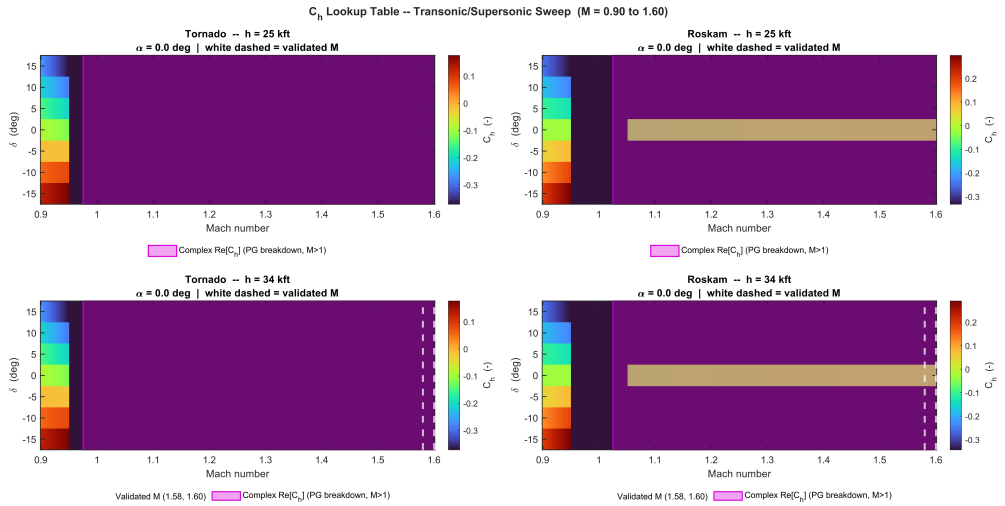


Figure 4.11: Hinge moment coefficient C_h as a function of Mach number and aileron deflection - transonic and supersonic sweep ($M = 0.90$–1.60). Magenta patches flag complex Roskam values ($M > 1$, PG breakdown). White dashed lines mark the two validated supersonic Mach numbers.

On the other hand the heatmap for higher Mach numbers conditions, demonstrate how both methods break approaching, and after, sonic conditions, yielding complex numbers in all cases except the Roskam predictions at zero deflections. The only reason for that happening is that the methodology gets computed directly with derivatives coefficients and angles ($C_{h_\delta}\delta$), then at null values of deflections the method is still able to predict a coefficient based on trim conditions.

4.6.3 Mach Sensitivity of C_h

Figure 4.12 shows the Mach sensitivity of C_h at a fixed angle of attack and aileron deflection, combining both extended sweeps to cover the full Mach range (0.50–1.60). Each altitude level is shown as a separate series; the boundary between the two sweeps are indicated by a vertical line.

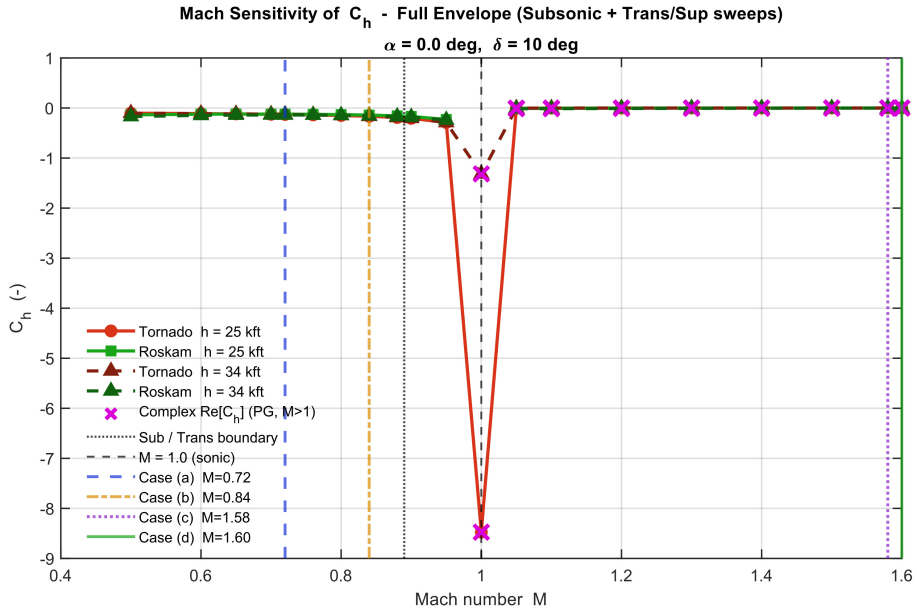


Figure 4.12: Hinge moment coefficient C_h as a function of Mach number for Tornado and Roskam, at fixed α and δ.

Once again the clear break is visible starting at sonic conditions after the result begin to deteriorate approaching $M = 1$. Before these condition is verified the consistency between the methods is highlighted, as well as the results being completely fixed to zero after sonic conditions are passed.

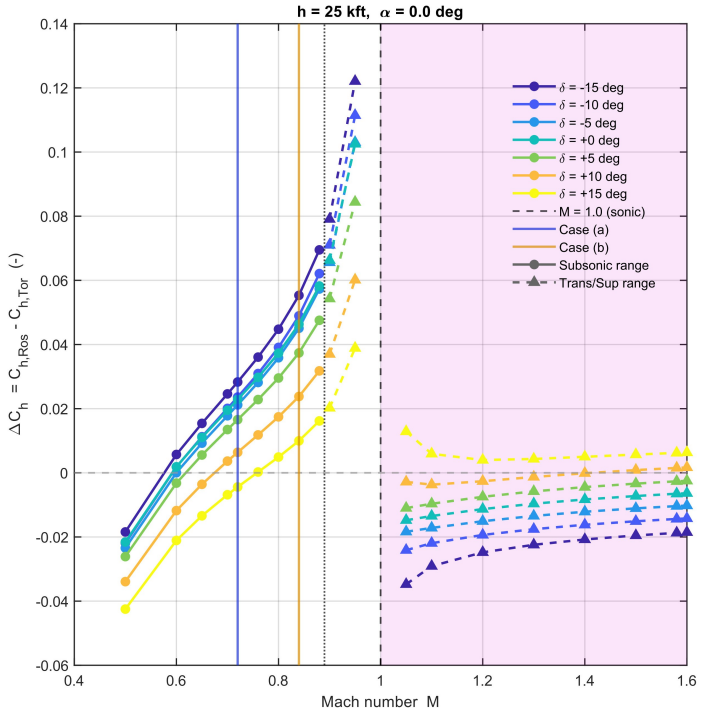
4.6.4 Roskam-Tornado Divergence

Figure 4.13 quantifies the absolute difference $\Delta C_h = C_{h,\text{Roskam}} - C_{h,\text{Tornado}}$ as a function of Mach number, for multiple fixed aileron deflections and both altitude levels. This representation makes the methodological divergence directly readable and identifies the Mach range over which the two methods can be considered equivalent for practical FCS sizing purposes.

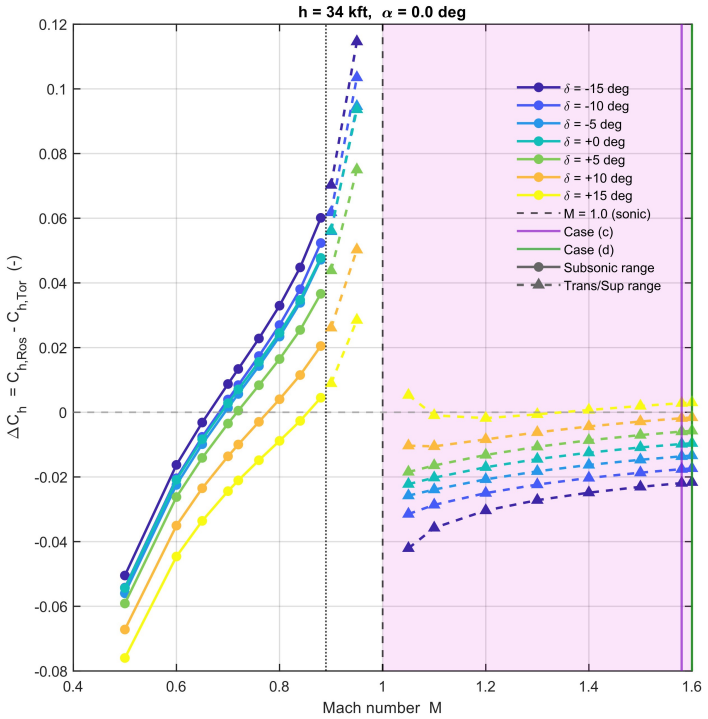
The plots show a clear divergence between the predictions from the two methodologies once sonic conditions are reached. The discrepancy is influenced by both Mach number and aileron deflection angle.

In the subsonic regime, the magnitude of the differences increases progressively with Mach number for all aileron deflections, with growth becoming more rapid as the transonic region is approached.

In the supersonic regime, the differences remain lower in magnitude and vary more smoothly with Mach number. Positive aileron deflections produce positive coefficient increments, while negative deflections produce negative increments.



(a) Absolute difference $\Delta C_h = C_{h,\text{Roskam}} - C_{h,\text{Tornado}}$ as a function of Mach number for several fixed deflection angles. The shaded region marks $M \geqslant 1$, where the Roskam PG correction is no longer physical.



(b) Absolute difference $\Delta C_h = C_{h,\text{Roskam}} - C_{h,\text{Tornado}}$ as a function of Mach number for several fixed deflection angles. The shaded region marks $M \geqslant 1$, where the Roskam PG correction is no longer physical.

Figure 4.13: Absolute difference $\Delta C_h = C_{h,\text{Roskam}} - C_{h,\text{Tornado}}$ as a function of Mach number for several fixed deflection angles.

5 Discussion

This chapter interprets the results presented in Chapter 4 in the context of the three research questions formulated in Section 1.3. Section 5.1 analyses the quantitative findings, relates them to the theoretical background of Chapter 2, and addresses each research question directly. Section 5.2 takes a self-critical stance on the methodology, examines the reliability, validity, and replicability of the study, and provides a critique of the sources used. The chapter concludes in Section 5.3 with a discussion of the broader picture of the work.

5.1 Results

5.1.1 Subsonic Prediction Accuracy and Method Comparison

For the two subsonic conditions, both methods produce estimates that are broadly consistent with the NASA flight data, and with the reference simulation of Chakraborty et al. [11]. The quantitative comparison in Table 4.3 shows that Roskam achieves a marginally lower RMS error than Tornado for both cases (a) and (b): 184.27 Nm versus 199.40 Nm at Mach 0.72, and 459.48 Nm versus 524.05 Nm at Mach 0.84. The absolute difference at Mach 0.72 is roughly 8%, which is modest but consistent throughout the maneuver duration.

This result is not unexpected given the theoretical foundations of the two methods. The Roskam semi-empirical formulation was calibrated against a broad experimental database and incorporates a Prandtl–Glauert compressibility correction that partially accounts for the subsonic Mach effects present at both conditions. Tornado, while also applying a Prandtl-Glauert correction by default in the current implementation (Section 4.1.2), remains fundamentally an incompressible vortex lattice solver: the correction scales the aerodynamic coefficients but does not alter the underlying potential-flow pressure distribution, which cannot represent the true compressible flow topology at Mach 0.72 and Mach 0.84. The residual advantage of Roskam is therefore attributable to the superior calibration of its empirical correction chain rather than to any fundamental difference in how compressibility is accounted for at the formulation level.

The residual time histories in Fig. 4.6 reveal that the error for both methods is not uniformly distributed across the maneuver, but is concentrated near peak deflection instants. This pattern is consistent with the breakdown of the linear hinge-moment model near maximum

deflection, where the aerodynamic loading becomes increasingly nonlinear, and the assumption of proportionality between C_h and δ is violated. The Roskam method, whose linear regime is specified to hold up to approximately $\delta \approx 20^\circ$ (Section 4.1.1), reaches this boundary precisely at the peaks of the maneuvers examined.

For case (a) at Mach 0.72, both methods perform similarly to the high-fidelity reference simulation (RMS: Chakraborty = 178.70 Nm, Roskam = 184.27 Nm, Tornado = 199.40 Nm), confirming that they could serve as viable tools for first-order hinge moment estimation at this flight condition. The gap relative to the reference simulation is modest and lies within the uncertainty band associated with the geometric assumptions and the best-alpha optimization procedure.

For case (b) at Mach 0.84, the absolute errors increase substantially for both methods, while the reference simulation retains a significantly lower RMS of 279.99 Nm against 459.48 Nm (Roskam) and 524.05 Nm (Tornado). This growing gap with increasing Mach number points directly to the role of compressibility: the high-fidelity simulation captures transonic effects that are only partially corrected by the Prandtl-Glauert factor in Roskam and are entirely absent in the incompressible Tornado formulation. Notably, the gap between Roskam and Tornado also narrows slightly from case (a) to case (b), from roughly 8% to roughly 12% in absolute terms, suggesting that the Prandtl-Glauert correction, while present in Roskam, is increasingly insufficient as the transonic regime is approached, and does not provide a growing advantage over Tornado's incompressible prediction.

The scatter correlation of Fig. 4.9 provides complementary evidence. Restricting the regression to the subsonic cases gives slopes of 0.991 for Tornado and 1.154 for Roskam: Tornado tracks the subsonic loads with few to no scale bias, whereas Roskam slightly overestimates their magnitude. Both lie close to the ideal 1:1 line, consistent with the comparable RMS errors of Table 4.3 and confirming that, within the subsonic regime, the two methods track the flight signal with similar fidelity (Fig. 4.5). The supersonic cases were excluded from the fit, since including them would drive the regression toward a near-zero slope dominated by the failure mode rather than the subsonic performance the metric is meant to characterize.

5.1.2 Supersonic Regime and Method Breakdown

Cases (c) and (d), flown at Mach 1.58 and Mach 1.60, expose the fundamental validity boundaries of both methods. Both the Roskam and the Tornado implementations apply a Prandtl–Glauert correction of the form $1/\sqrt{1-M^2}$, which becomes physically undefined for $M > 1$ as the factor yields imaginary values. The results presented in Section 4.4 adopt the real part of the complex C_h as a formal continuation of both estimates; however, this quantity has no direct aerodynamic interpretation in the supersonic regime and should be understood as an artifact of applying a subsonic correction beyond its domain of validity. Any numerical proximity to the flight data under these conditions would be accidental rather than physically meaningful.

Beyond the shared Prandtl–Glauert failure, the two methods differ in the structural nature of their breakdown. Tornado's incompressible vortex lattice formulation cannot represent the wave drag and shock-induced separation that dominate supersonic pressure distributions, even if the Prandtl–Glauert correction were replaced or disabled. Roskam's empirical chain was similarly calibrated on subsonic experimental data and carries no physical model of supersonic aerodynamics beyond the correction factor. The large RMS values of approximately 3300 Nm (Tornado) and 3600 Nm (Roskam) reported in Table 4.4 are fully consistent with this expectation. An interesting direction for future study would be to deactivate the Prandtl–Glauert correction in Tornado and replace it with a linearized supersonic correction, to assess

whether the VLM pressure integration can provide useful estimates when fed a physically appropriate compressibility factor, without requiring a full supersonic panel method.
The two supersonic Mach numbers are very close (1.58 and 1.60), and both methods produce nearly identical outputs for both cases. The fact that the NASA flight data for cases (c) and (d) differ observably, both in magnitude and in the time scale of the maneuver, with case (d) extending over several hundred seconds, is itself evidence of a supersonic sensitivity that neither method captures.

For fighter aircraft that usually operate at supersonic speeds, supplementing these tools with a transonic/supersonic correction would be a necessary extension for any practical FCS sizing application.

5.1.3 Effect of Flight-Condition Parameters on Accuracy

Mach number. The Mach number is the parameter that most clearly distinguishes the behavior of the two methods. As shown in Fig. 4.12, both methods apply a Prandtl-Glauert correction and therefore predict a C_h that grows in magnitude with Mach number through the subsonic regime; however, the empirical calibration chain in Roskam amplifies this effect more strongly than the direct coefficient scaling applied in Tornado. The absolute divergence between the two methods, quantified in Fig. 4.13, grows progressively as the transonic region is approached and increases with aileron deflection magnitude, making Mach number the dominant source of disagreement with the subsonic regime. Both methods become physically invalid supersonically due to the imaginary values produced by the shared Prandtl-Glauert factor; beyond that threshold, neither is applicable without augmentation. The physical expectation that compressibility amplifies aerodynamic loading supports the Roskam prediction as more representative than Tornado's within the subsonic regime, given its stronger empirical grounding, but neither can be trusted approaching Mach 1.

Aileron deflection. Aileron deflection angle is the primary driver of hinge moment magnitude in both methods and is the variable along which the lookup table is most densely sampled. The non-dimensional C_h versus δ plots in Fig. 4.8 show that both methods produce a near-linear relationship over the range $\delta \in [-15^\circ, 15^\circ]$, consistent with the theoretical $C_{h\delta}$ coefficient being approximately constant in this regime. The slopes agree between the two methods in the subsonic cases, which explains why the time-domain comparison in Fig. 4.5 is qualitatively similar for both methodologies. At the extremes of the deflection range ($|\delta| > 15^\circ$–20°), both methods diverge more strongly from the flight data, consistent with the onset of nonlinear aerodynamic effects and the known linear validity boundaries of both formulations.

Angle of attack. The angle-of-attack sensitivity analysis (Section 4.3.1 and Fig. 4.4) reveals that the optimal representative α for minimizing the reconstruction error is negative for both subsonic cases: -3° for case (a) and -5° for case (b). These values are somewhat lower than the nominal trim angle of attack expected at the respective flight conditions, which may reflect the effect of roll kinematics on the effective incidence of the inboard wing panel, or the fact that the static lookup table does not capture the $\dot{\alpha}$ contribution to the hinge moment during dynamic maneuvers. The RMS error curves are relatively shallow near their minima, particularly for case (a), indicating that the reconstruction is not highly sensitive to the exact value of α within a range of approximately $\pm 2^\circ$ around the optimum. This might also be a favorable property for a conceptual design tool where the precise trim angle of attack may not be known accurately.

Altitude. Altitude enters the hinge moment estimate only through the dynamic pressure $q_\infty = \frac{1}{2}\rho(h)\,(M\,a(h))^2$, which scales the dimensional M_h from the non-dimensional coef-

ficient C_h. Both $C_{h\alpha}$ and $C_{h\delta}$ are aerodynamic coefficients that are theoretically altitude-independent; their computation in the Roskam method relies only on geometry and Mach number, and the Tornado VLM formulation is inherently incompressible. The heatmaps of Fig. 4.10 confirm this expectation: the C_h distributions at 25 kft and 34 kft are structurally similar, with differences related also to the slightly different best-fit α used for each altitude slice.

5.1.4 Framework Usability and Efficiency

The automated framework addresses RQ1 by combining the Roskam semi-empirical method and the Tornado panel method in a shared parametric loop that sweeps the full flight envelope and stores results in a structured lookup table.

The lookup table architecture decouples the estimation step from the maneuver reconstruction, avoiding repeated solver calls during dynamic simulation. The pre-computed table (Fig. 2.5) enables the hinge moment early estimation at any operating point within the swept envelope to be retrieved in a single interpolation step, as demonstrated in Section 4.3.2. This is particularly valuable in the context of FCS sizing iterations, where the hinge moment must be evaluated for a large number of candidate maneuvers without re-executing the underlying solvers. The extended parametric sweeps of Section 4.6, approximately 486 and 462 cases respectively, each completed in approximately 15 minutes on a standard workstation, confirming that the framework is compatible with the rapid iteration cycles of conceptual design.

The CPACS-based interface establishes a standardized geometric input format consistent with established MDO environments [12]. The current implementation relies on a tool-specific extension rather than full schema compliance; this is a practical limitation but not a fundamental one, as the schema can be extended progressively as the tool matures. A fully compliant interface would enable the framework to be embedded directly in an MDO workflow driven by tools such as Desmo [15], without manual data conversion.

The modular separation of the Roskam and Tornado components means that either can be replaced or updated independently without affecting the shared infrastructure. A higher-fidelity panel method, such as a full-potential solver with compressibility corrections, could be substituted for Tornado with minimal integration effort, providing a clear upgrade path as design maturity increases.

5.2 Method

5.2.1 Modeling Assumptions and Their Consequences

Both methods are fundamentally limited to the attached-flow, linear aerodynamic regime. The linear superposition of $C_{h\alpha}$ and $C_{h\delta}$ implied by Eq. 2.1 is a first-order approximation that neglects nonlinear interactions between angle of attack and deflection, dynamic effects, and viscous contributions. For the aggressive maneuvering conditions typical of fighter aircraft, large deflections, high angles of attack, and transonic flight, the predictions should be treated as first-order estimates. This is not just a limitation of the current implementation but an inherent feature of both theoretical frameworks, as discussed in the background chapter (Sections 2.4 and 2.5).

The absence of viscous modeling in Tornado is a more specific concern for hinge moment estimation than for overall lift or pitching moment, because the pressure distribution near the hinge line is strongly influenced by boundary layer growth and trailing edge separation, both viscous phenomena. A semi-empirical viscous correction, such as those described

in [4], could improve Tornado's fidelity without sacrificing its computational efficiency and represents a well-defined near-term development path.

The static evaluation approach, which computes $C_{h\alpha}$ and $C_{h\delta}$ once per flight condition and applies them linearly across a dynamic maneuver, introduces an additional approximation. The dynamic hinge moment in a rolling maneuver is influenced by rate-dependent terms ($\dot{\alpha}$, $\dot{\delta}$) and lag effects that neither method captures. The best-alpha optimization procedure (Section 4.3.1) partially compensates for this by identifying a representative trim angle that minimizes the integrated reconstruction error, but this is a data-driven correction rather than a physics-based one, as discussed further in Section 5.2.2 below.

5.2.2 Validation Scope and Circularity

The validation is based on a single aircraft type (F/A-18 SRA) at two primary subsonic flight conditions. While these conditions are well-documented and constitute a rigorous first test, they do not cover the full spectrum of planform shapes, aileron chord ratios, or flight regimes relevant to conceptual design. In particular, highly swept delta wings, all-moving surfaces, and configurations with non-standard hinge line positions are not represented in the current validation set. The framework's accuracy for such geometries cannot be inferred from the present results, and broader validation remains a necessary step before the tool can be applied with confidence to arbitrary fighter concepts. The best-alpha optimization procedure further introduces a degree of circularity: the representative angle of attack is chosen to minimize error against the same flight data used to assess accuracy. This means the reported RMS values are optimistic estimates of out-of-sample prediction error. A genuinely blind validation, in which the representative α is determined from an independent model of the trim condition or from a separate dataset, would provide a more conservative and credible accuracy bound. Acknowledging this limitation is important because the framework is intended for configurations where no prior flight data exist; therefore, to fully validate the tool, the best-alpha selection cannot be performed against reference measurements.

5.2.3 Digitization Uncertainty

The empirical correction charts from Roskam [4] were digitized in MATLAB to enable automated evaluation (Section 3.1). This introduces a digitization uncertainty that was not formally quantified in this work. The magnitude of this uncertainty depends on the resolution of the original charts and the density of the digitized point cloud. A formal uncertainty-propagation study would quantify the contribution of digitization error to the overall prediction uncertainty and should be recommended for future development. The Roskam charts are also based predominantly on wind-tunnel data from the 1960s–1970s and may not fully represent modern low-observable airfoil profiles, introducing a further source of systematic bias that is difficult to quantify without additional experimental references.

5.2.4 Reliability, Validity, and Replicability

The reliability of the framework, understood as its ability to produce consistent results when applied repeatedly under the same conditions, is high. The lookup table generation is fully automated and deterministic given fixed inputs, and the only significant source of run-to-run variability is the numerical precision of the Tornado solver, which is negligible for well-resolved meshes. The digitized Roskam charts introduce a fixed systematic offset rather than random variability and therefore do not reduce reliability in the statistical sense.
Validity, the degree to which the computed hinge moments represent the true aerodynamic loading, is bounded by the modeling assumptions described above. Within the linear, attached-flow regime and the subsonic Mach range, the results of Section 4.3 confirm that

both methods have acceptable validity for conceptual design purposes, with RMS errors of 184-199 Nm at Mach 0.72 compared to a peak hinge moment of approximately 2000 Nm. Validity degrades predictably outside this envelope, as evidenced by the structurally large errors in Table 4.4 for the supersonic cases.
Replicability requires that a reader following the method description in Sections 3.1 and 3.2 can reproduce the lookup table from the same geometric inputs and flight conditions. The framework's automation and the public availability of both the Roskam methodology [4] and the Tornado solver [10] support this goal. The main obstacle to full replicability is implementation-specific digitized chart data; publishing the digitized datasets alongside the framework code would remove this barrier and is recommended before any open-source release.

5.2.5 Source Criticism

The primary sources used in this work can be assessed as follows. The NASA flight data of Navarro [13] constitute a technical report from a credible institutional source with documented instrumentation and calibration procedures; these data are treated as the ground-truth reference, and their reliability is considered high. The reference simulation of Chakraborty et al. [11] is a peer-reviewed conference contribution; its results are internally consistent with the Navarro data at subsonic conditions and are used appropriately as a comparative benchmark rather than as an independent ground truth, given that the simulation model itself embeds aerodynamic assumptions. The Roskam methodology [4] is a widely used industry reference with established credibility within its stated domain; the principal caveat, noted above, is that its empirical database reflects aircraft technology of several decades ago. The Tornado solver [10] is a validated, open-source VLM whose limitations are well documented in the literature and are consistent with those observed in this study; its use is appropriate for the targeted accuracy level.

5.3 The Work in a Wider Context

Although this thesis presents a methodological contribution to conceptual aircraft design and does not raise direct ethical concerns, several broader considerations are worth noting.

Dual-use nature of fighter design tools. The framework is intended for fighter aircraft, which are inherently military systems. By reducing time and cost during early design stages, such tools could indirectly accelerate combat aircraft development. This dual-use potential highlights the importance of responsible divulgation rather than diminishing the work's academic value.

Uncertainty communication and responsible use. Conceptual-design tools must clearly communicate their accuracy limits to prevent over-reliance on preliminary estimates. The RMS and MAE values reported in Tables 4.3 and 4.4, together with the validity boundaries identified in Section 5.2, are intended to indicate when higher-fidelity methods become necessary. Ignoring these limitations could lead to inappropriate actuator sizing decisions in a safety-critical system.

Open-source contribution and transparency. The use of open-source tools (Tornado, CPACS) and the reproducible nature of the framework help lower barriers to aircraft conceptual design research. Publishing the digitized Roskam charts and framework code under an appropriate license would further support transparency, peer review, and open-science practices.

6 Conclusion

6.1 Summary of Work

This thesis developed a parametric, modular framework for estimating aileron hinge moments during the conceptual design phase of fighter aircraft. The work was motivated by a methodological gap between low-fidelity empirical tools, which lack geometric sensitivity and compressibility fidelity beyond the linear regime, and high-fidelity CFD approaches, whose computational cost makes them unsuitable for the iterative evaluation cycles of early-stage design. The proposed framework bridges this gap by combining two complementary estimation methods within a shared, automated infrastructure.

The framework consists of three main components. First, an automated MATLAB implementation of Roskam's semi-empirical hinge moment methodology was developed, digitizing the relevant empirical correction charts and enabling rapid estimation across user-defined geometries and flight conditions without manual chart reading. Second, the Tornado vortex lattice method was adapted and integrated into the same input structure, providing a mid-fidelity panel-method alternative based on discrete aerodynamic force integration about the geometrically defined hinge axis. Third, a CPACS-based data exchange interface was implemented, enabling standardized communication between the geometry definition and both aerodynamic solvers, and laying the groundwork for future integration with multidisciplinary design environments such as Desmo [15].

Both methods were applied to a validation case based on the F/A-18 Systems Research Aircraft, using NASA experimental flight-test data [13] and the reference simulation of Chakraborty et al. [11] as benchmarks across four flight conditions spanning Mach 0.72 to Mach 1.60. The framework produced lookup tables covering approximately 1000 cases for the primary validation sweep and a further combined 948 cases across two extended parametric sweeps of the subsonic-to-supersonic Mach envelope.

6.2 Answers to the Research Questions

The three research questions posed in Section 1.3 can now be answered based on the results and discussion.

6.2.1 RQ1 - Framework development

Can a computationally efficient framework combining semi-empirical and panel methods provide sufficient accuracy for hinge moment estimation at the conceptual design stage?

The developed framework demonstrates that a computationally efficient solution is achievable at the conceptual design stage by combining an automated Roskam module, an adapted Tornado solver, and a precomputed lookup table architecture. The full primary validation sweep of approximately 1000 flight conditions completes in a timeframe compatible with iterative design practice, and the extended parametric sweeps of roughly 486 and 462 cases each required approximately 15 minutes on a standard workstation. The CPACS interface provides a standardized data-exchange pathway that positions the framework for embedding in broader multidisciplinary design workflows. The aim of providing a reusable, geometry-driven estimation tool for early FCS sizing has therefore been achieved within the defined scope.

6.2.2 RQ2 - Panel Method and Roskam Comparison

Does the integration of a panel method solver improve hinge moment prediction compared to a traditional empirical approach, such as Roskam? And how do they compare?

The integration of the Tornado panel method does not systematically improve hinge-moment prediction relative to Roskam for the subsonic conditions examined. Roskam achieves a marginally lower RMS error at both Mach 0.72 (184.27 Nm versus 199.40 Nm) and Mach 0.84 (459.48 Nm versus 524.05 Nm), primarily because its Prandtl-Glauert correction captures compressibility effects that the incompressible Tornado formulation neglects. At the same time, Tornado provides a geometrically consistent, panel-wise representation of the control-surface pressure distribution that is independent of the empirical calibration database underlying Roskam, making it the more transferable choice for unconventional planform or configurations outside the Roskam validity range. In the supersonic regime, both methods fail structurally for distinct but equally fundamental reasons. The two approaches are therefore best treated as complementary rather than substitutive tools at the conceptual design stage.

6.2.3 RQ3 - Flight Condition Parameters Effect ON Results Accuracy

Which flight condition parameters most significantly influence hinge moment prediction accuracy, and do the semi-empirical and panel methods involved respond to them differently?

The Mach number and the aileron deflection angle are the parameters that most significantly affect the accuracy of both methods. The Mach-number effect is particularly consequential because the two methods respond to it in fundamentally different ways: Roskam's Prandtl-Glauert correction amplifies the predicted C_h with increasing Mach, whereas Tornado's incompressible formulation produces a Mach-independent coefficient throughout. This divergence becomes the dominant source of inter-method disagreement as the transonic regime is approached, rendering both methods invalid in the supersonic regime. Aileron deflection is the primary driver of hinge moment magnitude and governs the onset of nonlinear aerodynamic effects beyond approximately 20°. Angle of attack has a secondary influence that is most practically significant in selecting the representative static α for dynamic reconstruction;

the shallow RMS minima observed in the sensitivity analysis indicate that the framework is tolerant of $\pm 2^\circ$ uncertainty in this parameter, a favorable property for conceptual design use. Altitude has no direct effect on the non-dimensional coefficient and influences only the dimensional scaling through dynamic pressure.

6.3 Key Findings

The principal findings of this work are as follows.
Both the Roskam and Tornado methods yield hinge-moment estimates that align with the NASA flight data and the high-fidelity reference simulation under subsonic conditions. At Mach 0.72, both methods produce RMS errors within approximately 12% of the reference simulation value of 178.70 Nm, confirming their viability as first-order estimation tools in the linear aerodynamic regime.

The accuracy of both methods degrades substantially as the Mach number approaches the transonic regime. At Mach 0.84, the RMS errors roughly double relative to Mach 0.72, and the gap relative to the reference simulation widens considerably, underscoring the growing importance of transonic effects that neither method can capture. In the supersonic regime, both methods yield RMS errors of 3300-3600 Nm, compared with measured hinge moments of approximately ± 6000 Nm, confirming that neither is applicable without augmentation beyond Mach 1.

The linear validity boundary for both methods coincides with the peak deflection amplitudes of the examined maneuvers ($|\delta| \approx 20^\circ$ in the subsonic cases), explaining the observed concentration of prediction error at peak deflection rather than during transition phases.

The lookup table architecture successfully decouples the estimation step from the maneuver reconstruction, enabling the dynamic hinge moment time history to be reproduced solely through table interpolation, without solver re-execution. This confirms the practical utility of the approach for FCS sizing workflows requiring repeated evaluation across candidate maneuvers.

The CPACS-based geometry interface, while currently relying on a tool-specific schema extension, demonstrates the feasibility of standardized data exchange between the geometry definition and the aerodynamic estimation modules, and establishes a clear pathway toward full MDO integration.

6.4 Implications for Practice and Research

For design engineers working at the conceptual phase of fighter aircraft development, the framework provides a validated, accessible first estimate of aileron hinge loads, generated directly from parametric geometry definitions without the need for CFD or wind-tunnel data. The reported accuracy bounds, RMS errors of approximately 184-199 Nm at Mach 0.72 against peak moments of the order of 2000 Nm, suggest that the framework is appropriate for preliminary actuator sizing and for identifying the flight conditions that drive peak loading, but should be supplemented by higher-fidelity methods before final actuator selection or structural sizing.

For researchers working on FCS load estimation, this work establishes a reproducible baseline by combining two widely available tools (Roskam and Tornado) within a CPACS-compatible infrastructure, serving as a reference implementation for comparison with future, more capable methods. The identified limitations, the absence of supersonic capability, the

lack of viscous corrections in Tornado, and the digitization uncertainty in the Roskam charts define a concrete development roadmap.

For the broader conceptual design community, the CPACS integration demonstrates that standardized aerodynamic data exchange is achievable with relatively modest implementation effort, thereby supporting the wider goal of multi-tool MDO workflows in which the hinge-moment framework constitutes one module among many.

7 Future Developments

The following development directions are identified and prioritized according to expected impact and implementation effort.

7.0.1 Supplementary Validation

The highest-priority extension is to apply the framework to additional fighter configurations, such as a more highly swept delta wing or an alternative aileron-to-wing chord ratio. This would establish whether the observed accuracy of the F/A-18 SRA is generalizable or geometry-specific, and help identify potential systematic biases in the empirical correction factors Roskam used for non-NACA profiles.

A formal propagation of digitization uncertainty in the Roskam charts through the full estimation chain should be performed. This would yield confidence intervals for the predicted C_h values and quantify the contribution of chart resolution to the modeling assumptions.

Systematic convergence analysis of the VLM discretization is required for future developments, with particular focus on panel density near the hinge line and aileron edges. The current mesh was defined without formal verification; identifying a minimum panel count for stable hinge-moment prediction would validate the setup and optimize the accuracy–cost trade-off.

7.0.2 Supersonic Correction and CPACS Libraries

A linearized supersonic correction should be implemented to verify the tool's estimation capabilities across the full flight envelope.
Full adoption of the official CPACS libraries would replace the current semi-customized schema extension, enabling automatic derivation of VLM partitioning parameters directly from native geometry definitions. This would eliminate manual pre-processing steps and improve interoperability within CPACS-based MDO workflows.

7.0.3 Additional Simulation Models and Tools

Integration of the hinge-moment database with a six-degree-of-freedom flight mechanics model, as outlined in Fig. 2.4, would enable trajectory-dependent evaluation of control surface loads. This would close the loop between aerodynamic loading and maneuver definition,

transforming the tool into an iterative FCS sizing framework.

Coupling the hinge-moment predictions with actuator models for EHA or EMA architectures [2] would enable early estimation of peak power demand and thermal loads. This capability directly supports electrical system sizing and power distribution design at the conceptual stage.

Full integration with Sizing Tools such as *Desmo* [15] as the overarching MDO environment would realize the workflow described in Section 1.4. Geometry would be imported via CPACS, aerodynamic loads computed by the present framework, and actuator sizing fed back into the design loop, enabling fully automated iterative FCS evaluation.

8 Appendices

8.1 Detailed Code Architecture

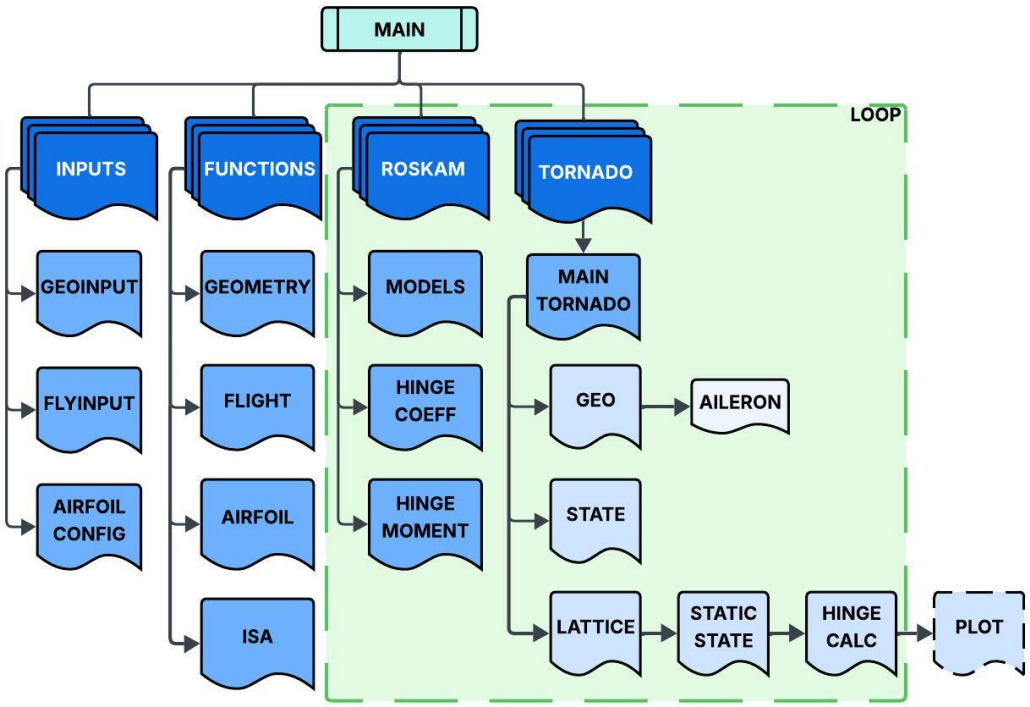


Figure 8.1: Detailed MATLAB code architecture

8.2 Input Tracking

Data	Description	XPath(s) in CPACS
Name	aircraft name	/cpacs/vehicles/aircraft/model/name
b	Wingspan	/cpacs/vehicles/aircraft/model/wings/wing/sections/section[last()]/transformation/translation/y (last section)
S	Wing Reference Area	/cpacs/vehicles/aircraft/model/reference/area
Lambda	Wing Sweep LE	atan2(x_tip_LE - x_root_LE, y_tip - y_root), x/y from /cpacs/vehicles/aircraft/model/wings/wing[@uID='wing']/sections/section/transformation/translation
c_r	Root chord	/cpacs/vehicles/aircraft/model/wings/wing/sections/section[1]/transformation/scaling/x
c_t	Tip Chord	/cpacs/vehicles/aircraft/model/wings/wing/sections/section[last()]/transformation/scaling/x

Figure 8.2: Example of Input Tracking Excel File

8.3 XSDM Diagrams

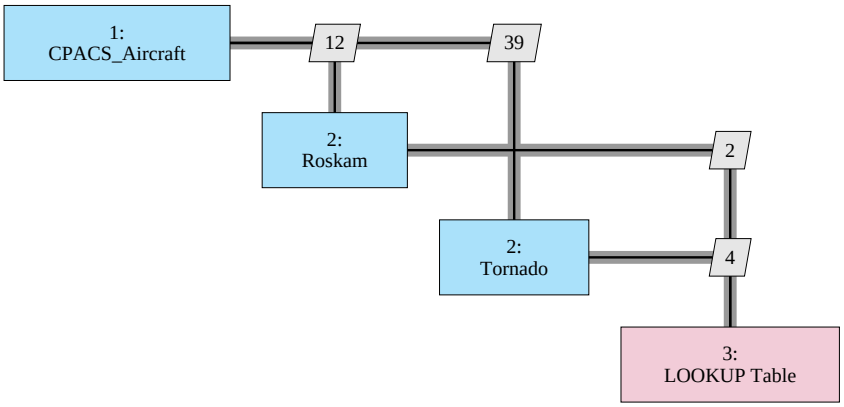


Figure 8.3: Full Context XSDM

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